CSS MERCANTILE LAW

T8KB

The Knowledge Base

Topic 8 · Electronic Transactions and Digital Evidence — taught from zero in plain English

WHAT THIS DOCUMENT GIVES YOU

- Electronic transactions and digital evidence from zero in plain English, read against the official texts of the Electronic Transactions Ordinance 2002, the Qanun-e-Shahadat Order 1984 (with the 2002 Schedule and the 2023 Article 164) and the Prevention of Electronic Crimes Act 2016
- One running example — Sara Fabrics, its Dubai buyer and an accredited certificate provider — through contract, signature, hacking and trial
- Twelve chapters from definitions, recognition and the signature presumption to dispatch and receipt, certification service providers, the Certification Council, offences, the evidence amendments, the leading digital-evidence cases and a critical assessment
- Three debates, the examiner's record and five likely questions
- 102 one-liners with every past MCQ, and a revision sheet

Federal Public Service Commission · CSS Competitive Examination · Optional paper, 100 marks

Written for CE-2027 onwards · Record: CE-2016 to CE-2026 · Facts current to October 2026

Contents

HOW TO BUILD THE CONTENTS

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If it does not ask: right-click the grey contents area → Update Field → Update entire table → OK. On a Mac, Control-click the same area. Or press Ctrl + A, then F9 (Windows), to update every field at once. Repeat after any edit, before you export to PDF.

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How to Use This Document

What this topic covers

Topic 8 is **Head VI — the Electronic Transactions Ordinance 2002** (Ordinance No. LI of 2002, 11 September 2002). The Ordinance makes **electronic documents and signatures** legally valid, sets rules for **when and where** electronic messages are sent and received, creates a system of **certification service providers** supervised by the **Certification Council**, punishes misuse, and amends the **Qanun-e-Shahadat Order 1984** so that electronic records can be proved in court. Since 2026 the examiner has also asked about **digital evidence** in Pakistan's courts.

The topic has had **7 visits** (2022–2026) and **4** of the recovered MCQs — all of them since 2022, which makes it the **newest** and fastest-growing topic in the paper. The examiner's favourites are **recognition and presumptions** (2023, 2024, 2026) and the **Certification Council** (2022, 2024); 2025 asked for a critical evaluation of the whole Ordinance, and 2026 asked about **electronic evidence** under the Qanun-e-Shahadat Order and the case of *Alamgir Khalid Chughtai v State*.

IN SIMPLE WORDS

Business used to run on **paper**: signed contracts, stamped deeds, original letters. Today it runs on **emails, PDFs, WhatsApp messages and online forms**. The law had to answer simple questions: Is an email a "writing"? Is a typed name a "signature"? When is an online offer "received"? Can a printout be shown to a judge? The Electronic Transactions Ordinance answers them — mostly with "yes, on conditions" — and builds a system of trusted **digital certificates** to make online dealing safe.

Our running example — Sara Fabrics goes digital

EXAMPLE · MEET THE PARTIES

Sara Fabrics, a textile exporter in Faisalabad run by **Ms Sara**, sells cloth to **Gulf Garments**, a buyer in Dubai. They agree prices by **email**, sign contracts with **digital signatures** issued by an accredited certification service provider called **SecureSign**, and exchange delivery updates on **WhatsApp**. Sara's accountant, **Imran**, has access to the company's email. When a payment dispute reaches a court in Lahore, the judge must decide what the emails, signatures and messages prove. Every rule in this document is shown on these people.

How this document is built

1. **Part Zero** gives the words of the topic, the Ordinance on one page and the life of an electronic contract.
2. **Part One** has twelve chapters: e-commerce and the Ordinance (1); definitions (2); legal recognition (3); electronic signatures and the presumption (4); stamp duty, attestation and excluded instruments (5); attribution, acknowledgment, dispatch and receipt (6); certification service providers (7); the Certification Council (8); offences and jurisdiction (9); the Qanun-e-Shahadat Order (10); digital evidence in the courts (11); and a critical assessment (12).
3. **Parts Two to Five** give the debates, the record with five likely questions, the one-liners with every past MCQ, and the revision sheet.

EXAMINER'S EYE · ONE ORDINANCE, THREE OTHER LAWS

The Ordinance has **never been amended** in the official text (Pakistan Code, uploaded March 2026). But it now sits beside three other laws you must mention: the **Qanun-e-Shahadat Order 1984**, which the Ordinance itself amended (s. 29 and Schedule) and whose **Article 164** was replaced in **2023**; the **Prevention of Electronic Crimes Act 2016 (PECA)**, the main cyber-crime law, amended in 2023 and 2025; and the **Digital Nation Pakistan Act 2025**. Every section here was read in the **official texts**.

Part Zero — The ABC of Topic 8

0.1 The words of this topic

Table 1 · The key terms of Topic 8 (★ = define it in one sentence, with its section).

Term	Section	Simple meaning, with an example
Electronic document ★	s. 2(1)(m)	Documents, records, information, communications or transactions in electronic form — an email, a PDF, a text
Electronic signature ★	s. 2(1)(n)	Any letters, numbers, symbols, images or characters in electronic form applied with intent to authenticate or approve
Advanced electronic signature ★	s. 2(1)(d)	A signature unique to the signer, under his sole control and showing any later change — or one accredited by the Council
Information system	s. 2(1)(p)	Any electronic system for creating, sending, receiving, storing or processing information
Originator · addressee	s. 2(1)(t), (c)	The sender · the intended recipient (not an intermediary)
Authenticity ★	s. 2(1)(f)	Identification of, and attribution to, a particular person or system
Integrity ★	s. 2(1)(q)	Not tampered with, altered or modified since a particular time
Security procedure	s. 2(1)(x)	An agreed or reasonably secure method — passwords, codes, encryption — to establish authenticity or integrity
Certificate	s. 2(1)(h)	A certificate by a service provider confirming authenticity or integrity
Certification service provider	s. 17	A business that issues certificates; it may be accredited by the Council
Certification Council ★	s. 18	The Electronic Certification Accreditation Council — five members; accredits service providers
Repository	s. 23	The Council's public database of certificates
Automated information system	QSO Arts. 46-A, 73	A computer working without human intervention; its printouts may be primary evidence

0.2 The Ordinance on one page

Table 2 · The structure of the Electronic Transactions Ordinance 2002.

Chapter	What it covers
1 — ss. 1–2	Title, extent, definitions
2 — ss. 3–12	**Recognition and presumption**: electronic forms, writing, originals, retention, signatures, proof, presumption, stamp duty, attestation, certified copies
3 — ss. 13–16	**Electronic documents**: attribution, acknowledgment, time and place of dispatch and receipt, government use
4 — s. 17	**Certification service providers**
5 — ss. 18–27	**The Certification Council**: establishment, qualifications, funds, functions, repository, accreditation, practice statement, decisions
6 — ss. 28–30	Amendments to the Telecommunication Act 1996 and the **Qanun-e-Shahadat Order** (with the Schedule); extension of paper words to electronic forms
7 — ss. 31–33	Excluded instruments; acts outside Pakistan; overriding effect

Chapter	What it covers
8 — ss. 34–39	**Offences**: false information, false certificates, privacy, damage to information systems
9 — ss. 40–45	Network providers' liability; secret keys; rules and regulations

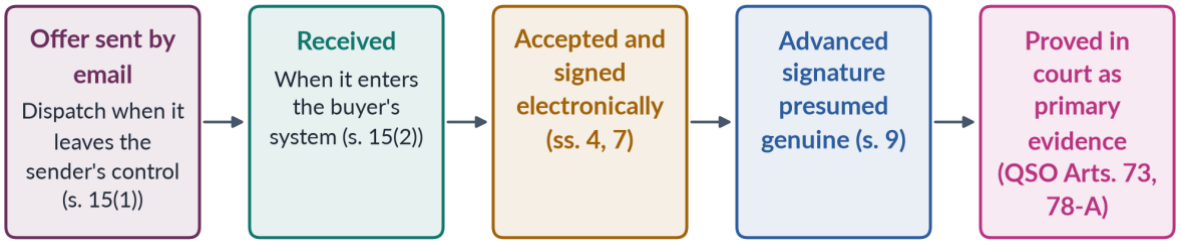


Figure 1 · The life of an electronic contract.

Chapter 1 · E-Commerce and the Electronic Transactions Ordinance 2002

*CSS 2025 Q7: "With the rise of digital communication and e-commerce, the Electronic Transactions Ordinance, 2002 has played a crucial role … Discuss the **key provisions** of the Ordinance …" MCQ 2026: "The Electronic Transactions Ordinance, 2002 primarily aims to … **All of these**" (legalise electronic documents, recognise electronic signatures, facilitate e-commerce).*

1.1 Why a law for electronic transactions was needed

IN SIMPLE WORDS · IN PLAIN WORDS

Old laws were written for **paper**. They said a contract "must be in writing", a deed "must be signed", a document "must be produced in original", a copy "must be certified". An email has no ink, a PDF has no original, and a typed name is not a pen signature. Without a new law, a businessman could never be sure that an online deal would stand up in court. The Ordinance removes that doubt by declaring that **electronic forms count**, on sensible conditions.

E-commerce means buying, selling and doing business by electronic means — online stores, email contracts, internet banking, electronic filing with government. For it to grow, the law must give three assurances:

1. **Validity** — an electronic contract or record is legally effective (ss. 3–7);
2. **Proof** — it can be proved in court, and its authorship and integrity can be relied on (ss. 8–9; Qanun-e-Shahadat Order);
3. **Trust** — there are trusted bodies that confirm who signed what (certification service providers and the Certification Council, ss. 17–27).

1.2 The Ordinance at a glance

- **Name and number.** The **Electronic Transactions Ordinance 2002**, Ordinance No. **LI of 2002**, promulgated on **11 September 2002** by the President during the period of the Proclamation of Emergency of 1999.
- **Purpose.** "To **recognize and facilitate** documents, records, information, communications and transactions in electronic form, and to provide for the **accreditation of certification service providers**" (long title).
- **Extent and commencement.** It "extends to the **whole of Pakistan**" and came into force **at once** (s. 1).
- **Continuing force.** Although made as an Ordinance, it appears in the official Pakistan Code as a law in force, and no amendment has been made to its text.



Figure 2 · The three aims of the Ordinance (MCQ 2026).

TRAP · MCQ 2026 Q14

The Ordinance aims to **legalise electronic documents**, **recognise electronic signatures** and **facilitate e-commerce** — the key is "**All of these**".

1.3 What the Ordinance covers — and what it does not

i) What it covers

Electronic **documents and records** (ss. 3–6), **signatures** (ss. 7–9), **contracts and communications** (ss. 13–15), **government** use of electronic forms (s. 16), **certification** (ss. 17–27), **evidence** (s. 29 and Schedule), **privacy of information systems** (s. 36) and related **offences** (ss. 34–37).

ii) What it does not cover

- **Excluded instruments** — negotiable instruments, powers of attorney, trusts (other than constructive, implied and resulting), wills and contracts for the sale or conveyance of **immovable property**, unless the Government extends the Ordinance to them (s. 31).
- **Intellectual property** in digital works — copyright and trademarks are governed by their own laws.

TRAP · MCQ 2025 Q11

"**Intellectual property rights in digital formats**" are **not** covered by the Ordinance. E-commerce contracts (ss. 3–4, 13–15), digital signatures (ss. 7–9) and privacy of information systems (s. 36) are.

1.4 The principles behind the Ordinance

i) Functional equivalence

The Ordinance asks what **function** a paper rule serves, and accepts an electronic form that serves the same function. "Writing" serves **later reference** — so an electronic document that is **accessible for subsequent reference** satisfies it (s. 4). An "original" serves **integrity** — so a record with **reliable assurance of integrity** satisfies it (s. 5).

ii) Non-discrimination

"No document … shall be **denied legal recognition** … on the ground that it is in **electronic form**" (s. 3).

iii) Technology neutrality

"Electronic" is defined widely — "electrical, digital, magnetic, optical, biometric, electro-chemical, wireless or electromagnetic technology" (s. 2(1)(l)) — so the law does not need changing with each new technology.

iv) Party autonomy

Most rules on attribution, acknowledgment, dispatch and receipt apply "**unless otherwise agreed**" between the parties (ss. 13–15).

v) A two-tier approach to signatures

Any **electronic signature** satisfies a legal signature requirement (s. 7), but an **advanced electronic signature** earns a statutory **presumption** of authenticity and integrity (s. 9).

These principles reflect the approach of the **UNCITRAL Model Laws on Electronic Commerce (1996) and on Electronic Signatures (2001)**, which many countries followed in the same period.

1.5 Scope in space and law

- **Acts outside Pakistan — s. 32.** The Ordinance applies "notwithstanding the matters … occurring **outside Pakistan**, in so far as they are directly or indirectly connected to, or have an effect on or bearing in relation to persons, information systems or events **within** the territorial jurisdiction of Pakistan". An email sent from Dubai to Sara Fabrics in Faisalabad is covered.
- **Overriding effect — s. 33.** The Ordinance applies "**notwithstanding anything to the contrary** contained in any other law".
- **Paper words extended — s. 30.** Words assuming paper — "attestation", "books", "document", "original", "record", "seal", "signature", "writing" and others — "shall, mutatis mutandis, **extend to electronic forms**".

1.6 The wider legal setting

Table 3 · The digital laws of Pakistan.

Law	Role
Electronic Transactions Ordinance 2002	Validity of electronic documents and signatures; certification; Qanun-e-Shahadat amendments
Qanun-e-Shahadat Order 1984	Law of evidence — amended by the 2002 Ordinance (Schedule) and by the Criminal Laws (Amendment) Act 2023 (Art. 164)
Payment Systems and Electronic Fund Transfers Act 2007	Electronic payments and consumer protection (Topic 5)
Prevention of Electronic Crimes Act 2016	Cyber-crimes, investigation, electronic evidence; amended 2023 and 2025
Digital Nation Pakistan Act 2025	National Digital Commission and Pakistan Digital Authority for digital governance

1.7 Why the Ordinance matters for ordinary people

The Ordinance is not only for large companies. A student who accepts an admission offer by email, a shopkeeper who takes orders on WhatsApp, a farmer who receives a payment notification on his phone, and a widow who files a pension claim online all rely on the idea that **electronic records count**. When disputes arise — a missing payment, a denied order, a forged message — the Ordinance and the evidence rules it introduced decide whether those records will be believed. Understanding the Ordinance is therefore understanding how trust works in the **digital economy**: who can be held to a message, how a signature is proved, and which bodies stand behind digital certificates.

REMEMBER · THE ORDINANCE IN ONE LINE

Ordinance LI of 2002 (11 September 2002) — recognises electronic documents, records and signatures (ss. 3–9), regulates communications (ss. 13–15), accredits certification service providers through the Certification Council (ss. 17–27), amends the Qanun-e-Shahadat Order (s. 29), excludes certain instruments (s. 31), reaches acts abroad with effect in Pakistan (s. 32), overrides other laws (s. 33) and punishes misuse (ss. 34–39).

Chapter 2 · The Key Definitions

*Syllabus: "Definitions". Many answers turn on a definition — especially **electronic signature**, **advanced electronic signature**, **authenticity**, **integrity** and **security procedure**. Learn them word for word.*

2.1 Electronic, information and information system

✦ THE LAW

Section 2(1)(l), (o), (p) · *Electronic Transactions Ordinance 2002*

(l) "**electronic**" includes electrical, digital, magnetic, optical, biometric, electro-chemical, wireless or electromagnetic technology; (o) "**information**" includes text, message, data, voice, sound, database, video, signals, software, computer programs, codes including object code and source code; (p) "**information system**" means an electronic system for creating, generating, sending, receiving, storing, reproducing, displaying, recording or processing information.

IN SIMPLE WORDS · IN PLAIN WORDS

"Electronic" covers almost every technology — a mobile phone, a fingerprint scanner, a wireless network. "Information" is anything a computer can hold — words, voice notes, videos, software. An "information system" is any machine or network that handles such information — a laptop, an email server, a bank's computer, a WhatsApp account.

2.2 Electronic document and transaction

- "**Electronic document**" includes "documents, records, information, communications or transactions in electronic form" (s. 2(1)(m)).
- "**Transaction**" means "an act or series of acts in relation to creation or performance of rights and obligations" (s. 2(1)(z)).
- "**Automated**" means "without active human intervention" (s. 2(1)(g)) — for instance, an online store that confirms orders automatically.

2.3 The parties to a communication

- "**Originator**" — the person by whom, or on whose behalf, an electronic document purports to have been generated or sent, but **not an intermediary** (s. 2(1)(t)).
- "**Addressee**" — the person intended by the originator to receive it, but **not an intermediary** (s. 2(1)(c)).
- "**Intermediary**" — a person acting as a **service provider** for sending, receiving, storing or processing the communication (s. 2(1)(r)) — an email provider or messaging platform.
- "**Network service provider**" — one who owns, operates or controls a public switched network or provides telecommunication services (s. 2(1)(s)).
- "**Person**" includes an individual, appropriate authority, trust, **waqf**, association, statutory body, firm, company, joint venture or consortium, or any other entity, registered or not (s. 2(1)(u)).

EXAMPLE · WHO IS WHO

Ms Sara emails an offer to Gulf Garments through Gmail. **Sara Fabrics** is the **originator**; **Gulf Garments** is the **addressee**; Google, which carries and stores the email, is an **intermediary**; the mobile company carrying the data is a **network service provider**.

2.4 The two kinds of electronic signature

✦ **THE LAW**

Section 2(1)(n) · *Electronic Transactions Ordinance 2002*

"**Electronic signature**" means any **letters, numbers, symbols, images, characters** or any combination thereof in electronic form, applied to, incorporated in or associated with an electronic document, **with the intention of authenticating or approving the same**, in order to establish **authenticity or integrity, or both**.

✦ **THE LAW**

Section 2(1)(d) · *Electronic Transactions Ordinance 2002*

"**Advanced electronic signature**" means an electronic signature which is either — (i) **unique** to the person signing it, **capable of identifying** such person, created in a manner or using a means under the **sole control** of the person using it, and attached to the electronic document in a manner that **any subsequent change** in the electronic document is **detectable**; or (ii) provided by an **accredited certification service provider** and **accredited by the Certification Council** as being capable of establishing authenticity and integrity of an electronic document.

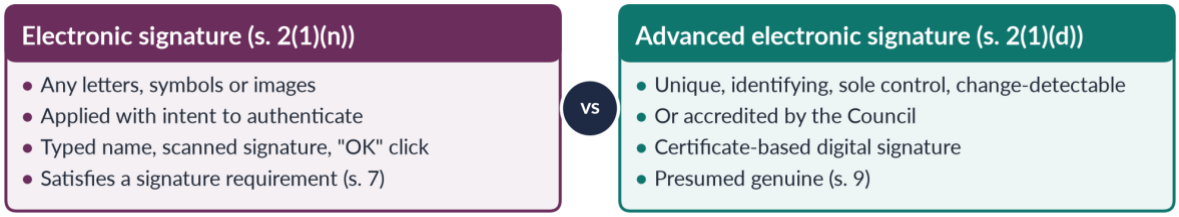


Figure 3 · Electronic and advanced electronic signatures.

EXAMPLE · THREE WAYS MS SARA SIGNS

(1) She types "Sara" at the end of an email — an **electronic signature**. (2) She pastes a scanned image of her pen signature into a PDF — also an **electronic signature**. (3) She signs the export contract with a **digital certificate** issued by **SecureSign**, an accredited provider, so that any later change to the PDF breaks the signature — an **advanced electronic signature**.

2.5 Authenticity, integrity and security procedure

- "**Authenticity**" — "the identification of and attribution to a particular person or information system" (s. 2(1)(f)): **who** made it.
- "**Integrity**" — the document or signature "has **not been tampered with, altered or modified** since a particular point in time" (s. 2(1)(q)): **whether it has changed**.
- "**Security procedure**" — a procedure that (i) is **agreed** between the parties, (ii) is implemented in the normal course by a business and is **reasonably secure and reliable**, or (iii) for a certificate, is specified in the provider's **certification practice statement** — for establishing authenticity or

integrity, and which may use algorithms, codes, identifying words and numbers, **encryption**, answer-back or acknowledgment procedures, software or hardware (s. 2(1)(x)).

2.6 The certification vocabulary

Table 4 · Certification terms (s. 2(1)).

Term	Clause	Meaning
Certificate	(h)	Issued by a certification service provider to confirm the authenticity or integrity of information, a document or a signature
Accredited Certification Service Provider	(b)	A provider accredited under the Ordinance to issue certificates for its cryptography services
Accreditation certificate	(a)	Granted by the Certification Council to a provider
Valid accreditation certificate	(aa)	One not suspended or revoked
Certification practice statement	(j)	The provider's statement of its practices in issuing certificates
Cryptography services	(k)	Transforming a document into a form that unauthorised persons cannot understand or decode
Repository	(w)	The information system for storing and retrieving certificates (s. 23)
Subscriber	(y)	A person who subscribes to a provider's services
Certification Council	(i)	The Electronic Certification Accreditation Council (s. 18)

2.7 The appropriate authority

"**Appropriate authority**" means the **Federal** Legislature or Government for Federal List matters; for Concurrent List matters with a Federal law in force, the Federal Legislature or Government, otherwise the **Provincial** Legislature or Government; a **statutory body** discharging government functions; and the **Supreme Court** or a **High Court** for their own procedural rules (s. 2(1)(e)). The term matters for **government e-services** (s. 16), **certified copies** (s. 12) and **attestation** (s. 11).

2.8 Using the definitions in an answer

Examiners reward answers that **apply** the definitions rather than merely list them. A good method is to ask three questions of any electronic record in a problem:

1. **Is it an electronic document?** Almost always yes — s. 2(1)(m) is wide enough to cover emails, files, messages and database entries.
2. **Who is the originator, and was an intermediary involved?** This leads into the attribution rules of s. 13 and the network provider's protection in s. 40.
3. **What kind of signature, and what security procedure?** An ordinary electronic signature needs proof (s. 8; QSO Art. 78-A); an advanced one earns the presumption (s. 9); the existence of a **security procedure** (s. 2(1)(x)) also decides whether a printout is primary evidence (QSO Art. 73, Explanation 4).

REMEMBER · THE DEFINITIONS IN ONE LINE

Electronic = any electro-magnetic, digital or biometric technology (s. 2(1)(l)); electronic document = documents, records, communications or transactions in electronic form (m); electronic signature = any electronic mark applied with intent to authenticate (n); advanced electronic signature = unique, identifying, under sole control and change-detectable, or Council-accredited (d); authenticity = attribution (f); integrity = unaltered (q); security procedure = agreed or reasonably secure method (x).

Chapter 3 · Legal Recognition — Writing, Originals and Retention

*CSS 2024 Q7: "Write a detailed note on **recognition and presumption** of electronic transactions." CSS 2023 Q5: "What considerations have been provided … for the purpose of **recognition and appraisal** of an electronic document?" Syllabus: "Recognition and presumption of electronic transactions (writing, signature, stamp duty, attestation etc.)".*

3.1 The basic rule — s. 3

✦ THE LAW

Section 3 · *Electronic Transactions Ordinance 2002*

No document, record, information, communication or transaction shall be denied legal recognition, admissibility, effect, validity, proof or enforceability on the ground that it is in electronic form and has not been attested by any witness.

IN SIMPLE WORDS · IN PLAIN WORDS

A deal is not invalid **just because** it was made on a computer or phone. An email contract, an online order, a WhatsApp confirmation — none of them can be thrown out **only** because it is electronic or has no witness. It may still fail for other reasons (no agreement, fraud, a missing requirement), but not for being electronic.

Section 3 works on **seven** fronts at once: legal **recognition**, **admissibility** in evidence, legal **effect**, **validity**, **proof** and **enforceability** — and it removes the objection that the document was **not attested** by a witness.

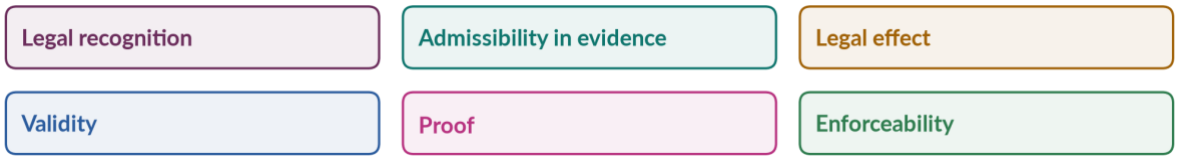


Figure 4 · What s. 3 protects an electronic document from losing.

3.2 Writing — s. 4

✦ THE LAW

Section 4 · *Electronic Transactions Ordinance 2002*

The requirement under any law for any document, record, information, communication or transaction to be **in written form** shall be deemed satisfied where the document … is in **electronic form**, **if the same is accessible so as to be usable for subsequent reference**.

- **The test is accessibility.** A paper writing can be read again later; an electronic document satisfies "writing" if it can be **opened and used later** — saved, stored and retrievable.
- **What fails.** A voice call that is not recorded, or a message set to vanish, is not "accessible for subsequent reference".
- **Example.** The Contract Act does not generally require writing, but where another law does — for instance, an arbitration agreement must be "written" (Topic 6) — an email containing the arbitration clause satisfies the requirement if it is saved and retrievable.

3.3 Original form — s. 5

✦ THE LAW

Section 5(1) · *Electronic Transactions Ordinance 2002*

The requirement under any law for any document … to be presented or retained in its **original form** shall be deemed satisfied … if: (a) there exists a **reliable assurance as to the integrity** thereof from the time when it was first generated in its **final form**; and (b) it is required that the presentation thereof is capable of being **displayed in a legible form**.

i) Integrity

The test is "whether the same has remained **complete and unaltered**, apart from the addition of any endorsement or any change which arises in the **normal course of communication, storage or display**" (s. 5(2)(a)).

ii) Reliability

The standard of reliability is "assessed having regard to the **purpose** for which the document … was generated and all other **relevant circumstances**" (s. 5(2)(b)).

IN SIMPLE WORDS · WHY "ORIGINAL" IS A PROBLEM ONLINE

Every copy of a file is identical to every other copy, so the idea of a single "original" does not fit. The Ordinance replaces it with **integrity**: if we can be reliably sure that the file has not changed since it was finalised — for example, because it carries an **advanced electronic signature** or was stored on a secure system — it counts as an original.

3.4 Retention — s. 6

Where any law requires a document to be **retained** — tax records, company books, bank records — retention in electronic form suffices if:

- (a) the contents remain **accessible** so as to be usable for subsequent reference;
- (b) the contents and form are **as originally generated, sent or received**, or can be shown to represent them **accurately**; and
- (c) information enabling identification of the **origin and destination** and the **date and time** of generation, sending or receipt is also retained.

EXAMPLE · SARA FABRICS KEEPS ITS RECORDS ONLINE

The tax authorities require Sara Fabrics to keep sales invoices. It stores them as PDFs on a secure server, with each email's **sender, recipient, date and time** preserved. The retention requirement is met (s. 6). If Imran instead printed the invoices, deleted the emails and kept only edited spreadsheets, condition (b) and (c) would fail.

3.5 Paper words extended — s. 30

"Notwithstanding anything contained in any other law", the expressions "attestation", "books", "books of accounts", "certificate", "charts", "deed", "document", "document of title", "execution", "instrument", "ledger", "map", "original", "plans", "publish", "record", "register", "seal", "signature", "witnessing", "words", "writing", and other words assuming paper, "shall, **mutatis mutandis**, **extend to electronic**

forms thereof" (s. 30). This gives the Ordinance's recognition rules a **wide reach** across the whole statute book.

3.6 Government and electronic forms — s. 16

- **No right to insist.** Nothing in the Ordinance gives anyone a **right** to require a government body to **accept or issue** electronic documents or payments (s. 16(1)).
- **Permission to go digital.** But any **appropriate authority** may, despite contrary laws, **accept filings**, **issue permits, licences and certificates**, and make **payments and procurement** in electronic form (s. 16(2)).
- **Standards.** It may specify the **format**, the **type of signature** or security procedure, the acceptable **certification service providers**, and **controls** for integrity, security and confidentiality (s. 16(3)).

3.7 The considerations for recognising and appraising an electronic document (2023)

Table 5 · The considerations the Ordinance applies to an electronic document.

Consideration	Section	Question the court asks
Non-discrimination	3	Is it being rejected only because it is electronic or unwitnessed?
Accessibility	4, 6(a)	Can it be opened and used for later reference?
Integrity	5(1)(a), 5(2)(a)	Has it remained complete and unaltered since it was finalised?
Reliability	5(2)(b)	Is the assurance reliable, given its purpose and circumstances?
Legibility	5(1)(b)	Can it be displayed in readable form?
Accuracy of form	6(b)	Is it as originally generated, sent or received?
Origin, destination, time	6(c)	Are sender, recipient, date and time preserved?
Signature	7–9	Is it signed electronically — and is the signature advanced?
Attribution	13	Whose communication is it?
Evidence rules	QSO Arts. 73, 78-A	Is the printout primary evidence? Must a security procedure be proved?

3.8 Recognition in practice

EXAMPLE · THE EMAIL THAT THE BUYER CALLS "JUST A COPY"

In court, Gulf Garments argues that Sara's printout of its acceptance email is only a "copy" and that no "original signed letter" exists. The judge applies s. 3 (no denial merely because electronic), s. 4 (the email is accessible for later reference, so "writing" is satisfied), and s. 5 (Sara's mail server shows the email unchanged since it was received, so the "original" requirement is met). The remaining question is **proof of authorship**, which turns on attribution (s. 13) and, if Gulf denies sending it, proof of the **security procedure** under QSO Article 78-A.

REMEMBER · RECOGNITION IN ONE LINE

No denial of recognition, admissibility, effect, validity, proof or enforceability merely because a document is electronic or unwitnessed (s. 3); writing = accessible for later reference (s. 4); original = reliable assurance of integrity plus legible display (s. 5); retention = accessible, accurate, with origin, destination and time (s. 6); paper words extend to electronic forms (s. 30); government may — but need not — go digital (s. 16).

Chapter 4 · Electronic Signatures and the Presumption

*CSS 2026 Q6: "**Critically examine the presumptions** … regarding the **authenticity and integrity** of electronic records and signatures. Analyze how these presumptions facilitate the reliability of electronic transactions and the **evidentiary challenges** that may arise in court." CSS 2024 Q7: "… recognition and **presumption** …". MCQs 2025 and 2026 on valid electronic and digital signatures.*

4.1 Recognition of electronic signatures — s. 7

✦ **THE LAW**

Section 7 · *Electronic Transactions Ordinance 2002*

The requirement under any law for **affixation of signatures** shall be deemed satisfied where **electronic signatures or advanced electronic signature** are applied.

IN SIMPLE WORDS · IN PLAIN WORDS

If a law says a document must be "signed", an **electronic signature** of any kind satisfies it — a typed name, a scanned signature, a click on "I agree", or a cryptographic digital signature. But not all signatures are equally strong in court: only an **advanced** electronic signature gets the law's automatic trust (s. 9).

TRAP · MCQS 2025 Q10 AND 2026 Q19

2026: "Any electronic method applied with intent to authenticate", "a digital signature generated through cryptographic means" and "a secure electronic signature verified by a certificate" are **all** valid electronic signatures under ss. 2(1)(n) and 7 — the key is "**All of these**". **2025:** the archive keyed "created using an **approved certification framework**" for what a digital signature is "required to be"; strictly, the Ordinance requires accreditation only for the **advanced** signature of s. 2(1)(d)(ii) — an ordinary electronic signature needs no certification.

4.2 Proof of electronic signature — s. 8

"An electronic signature **may be proved in any manner**, in order to verify that the electronic document is of the person that has executed it with the intention and for the purpose of verifying its **authenticity or integrity or both**" (s. 8). Proof may come from witnesses, the email account's records, IP logs, admissions, forensic examination or the provider's certificate.

4.3 The presumption — s. 9

✦ **THE LAW**

Section 9 · *Electronic Transactions Ordinance 2002*

In any proceedings, involving an **advanced electronic signature**, it shall be **presumed unless evidence to contrary is adduced**, that: (a) the electronic document affixed with an advanced electronic signature, as is the subject-matter of or identified in a **valid accreditation certificate**, is **authentic and has integrity**; or (b) the advanced electronic signature is the **signature of the person to whom it correlates**, the advanced electronic signature was **affixed by that person with the intention of signing or approving** the electronic document and the electronic document **has not been altered** since that point in time.

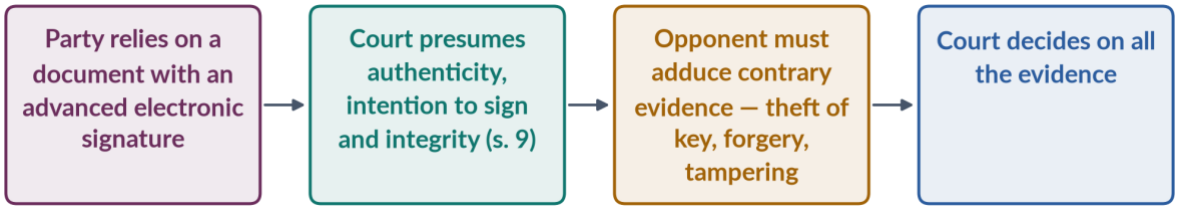


Figure 5 · How the s. 9 presumption works in court.

i) What is presumed

- **Authenticity** — the signature belongs to the person it correlates with;
- **Intention** — he affixed it **to sign or approve** the document;
- **Integrity** — the document **has not been altered** since signing.

ii) The nature of the presumption

It is **rebuttable**: "unless evidence to contrary is adduced". The burden of producing contrary evidence shifts to the party denying the signature.

iii) Only advanced signatures

The presumption is **not** available for an ordinary electronic signature. A typed "Sara" must be proved by evidence under s. 8; a certificate-based signature from **SecureSign** is presumed genuine.

4.4 The link with evidence law — QSO Article 78-A

The Ordinance inserted **Article 78-A** in the Qanun-e-Shahadat Order: "If an electronic document is alleged to be signed or to have been generated wholly or in part by any person through the use of an information system, and **where such allegation is denied**, the **application of a security procedure** to the signature or the electronic document **must be proved**." Read with s. 9, this means: an **advanced** signature is presumed genuine; any **other** disputed electronic signature or document requires proof that a **security procedure** (password, OTP, encryption, certificate) was applied.

EXAMPLE · THE DISPUTED CONTRACT

Gulf Garments denies signing a revised contract raising the price. (a) If Sara relies on a PDF signed with Gulf's **SecureSign** certificate, the court **presumes** the signature genuine and the document unaltered (s. 9); Gulf must prove, for instance, that its private key was stolen. (b) If Sara relies only on an email with Gulf's typed name, Gulf's **denial** means Sara must prove the **security procedure** — that the email came from Gulf's password-protected account (QSO Art. 78-A) — and may use any proof (s. 8).

4.5 How the presumptions make transactions reliable (2026)

- **Certainty.** Parties using accredited signatures know their documents will be **trusted** in court without expert evidence in every case.
- **Lower cost of proof.** The burden moves to the person who **denies** — usually the one with knowledge of any misuse.
- **Incentive to use secure technology.** Because only **advanced** signatures earn the presumption, businesses are encouraged to adopt **certificate-based** signatures.

- **Integration with evidence law.** Printouts of automated systems are **primary evidence**, with the system **presumed in working order** (QSO Art. 73, Explanation 3), and certificates in the **repository** are **public documents** (Art. 85(6)).

4.6 The evidentiary challenges (2026)

- **Key compromise.** If a private key or token is stolen, the presumption works **against the true owner**, who must prove the misuse — difficult without forensic records.
- **Low uptake.** Few people in Pakistan hold accredited certificates, so most electronic documents — WhatsApp messages, emails, scans — get **no presumption** and need full proof.
- **Proving security procedures.** Under Art. 78-A, a party must explain passwords, OTPs or system logs — technical evidence many courts and lawyers find hard to handle.
- **Tampering and deepfakes.** Screenshots and recordings are easily edited; courts insist on **forensic** verification of audio and video (Chapter 11).
- **The working-order presumption.** The presumption that an automated system was working (Art. 73, Expl. 3) may be hard for an individual to rebut against a bank or telecom company.
- **Rebuttal standard.** The Ordinance does not say how strong the "evidence to contrary" must be — leaving it to judicial discretion.

BALANCED VIEW · ARE THE PRESUMPTIONS WELL DESIGNED?

For: they follow the logic of the s. 118 presumptions for negotiable instruments (Topic 5) — trust the instrument, let the denier prove otherwise — and reward secure technology. **Against:** they assume a working certification ecosystem that has developed slowly, and they may burden victims of identity theft. **The balance:** keep the presumption for **accredited advanced** signatures, but courts should accept credible evidence of compromise readily, and the law should require providers to keep **audit trails** that help signers rebut misuse.

4.7 Rebutting the presumption in practice

A party who denies an advanced electronic signature must "adduce" contrary evidence (s. 9). Typical evidence includes: proof that the signer's **token or key was stolen** and reported to the provider before the signing; **device logs** showing that the signing took place from an unfamiliar computer or location; proof that the provider's **certificate was suspended or revoked** in the repository at the time; or forensic evidence that the document was **altered** after signing. Once such evidence is produced, the court weighs the whole evidence without the presumption.

REMEMBER · SIGNATURES IN ONE LINE

Any electronic signature satisfies a legal requirement (s. 7) and may be proved in any manner (s. 8); an advanced electronic signature is presumed authentic, intended and unaltered unless contrary evidence is adduced (s. 9); a denied electronic document or signature requires proof of a security procedure (QSO Art. 78-A).

Chapter 5 · Stamp Duty, Attestation, Certified Copies and Excluded Instruments

*Syllabus: "Recognition and presumption of electronic transactions (writing, signature, **stamp duty, attestation** etc.)". CSS 2024 Q7 and 2025 Q7 (key provisions). These rules complete the "recognition" answer.*

5.1 Why these rules were needed

IN SIMPLE WORDS · IN PLAIN WORDS

Many paper documents need more than a signature. Some must carry **stamp paper** (stamp duty), some must be **witnessed** or **notarised** (attestation), and courts often want **certified copies**. In 2002 there was no way to pay stamp duty on an email or to have a notary attest a PDF. The Ordinance gave a temporary solution for the first two and a method for the third — and it kept some especially important documents on paper altogether.

5.2 Stamp duty — s. 10

✦ **THE LAW**

Section 10 · *Electronic Transactions Ordinance 2002*

Notwithstanding anything contained in the **Stamp Act, 1899**, for a period of **two years** from the date of commencement of this Ordinance **or till the time the Provincial Governments devise and implement appropriate measures** for payment and recovery of stamp duty through electronic means, **whichever is later**, stamp duty **shall not be payable** in respect of any instrument executed in electronic form.

- **The rule.** No stamp duty on electronic instruments for **two years**, **or until** the Provinces put in place **electronic payment** of stamp duty — **whichever is later**.
- **Why "whichever is later".** The exemption continues as long as the Provinces have **not** devised electronic means; once they do, duty becomes payable.
- **Why the Provinces.** Stamp duty on most instruments is a **provincial** revenue; the Ordinance leaves it to the Provinces to create systems. Since then, several Provinces have introduced **e-stamping** systems for paper instruments, which change the practical position.

5.3 Attestation and notarisation — s. 11

✦ **THE LAW**

Section 11 · *Electronic Transactions Ordinance 2002*

Notwithstanding anything contained in any law for the time being in force, **no electronic document shall require attestation and notarization** for a period of **two years** from the date of commencement of this Ordinance **or till the time the appropriate authority devise and implement measures** for attestation and notarization of electronic documents, **whichever is later**.

- **The rule.** Electronic documents need **no attestation or notarisation** until the **appropriate authority** creates measures for electronic attestation — with a minimum period of two years.
- **Link with s. 3.** Section 3 already says no electronic document shall be denied recognition because it "has not been **attested by any witness**".

- **Link with s. 30.** "Attestation" and "witnessing" extend to electronic forms, so an appropriate authority may provide for **electronic attestation**.

5.4 Certified copies — s. 12

"Where any law requires or permits the production of **certified copies** of any records, such requirement or permission shall extend to **printouts or other forms of display** of electronic documents where, in addition to fulfilment of the requirements … specified in such law relating to certification, it is **verified in the manner laid down by the appropriate authority**" (s. 12). A court or registry may therefore issue certified **printouts** of its electronic records, once the authority lays down the verification method.

5.5 Excluded instruments — s. 31

✦ THE LAW

Section 31(1) · *Electronic Transactions Ordinance 2002*

Subject to sub-section (2), nothing in this Ordinance shall apply to: (a) a **negotiable instrument** as defined in section 13 of the Negotiable Instruments Act, 1881; (b) a **power-of-attorney** under the Powers of Attorney Act, 1882; (c) a **trust** … but excluding constructive, implied and resulting trusts; (d) a **will** or any form of testamentary disposition; and (e) a **contract for sale or conveyance of immovable property** or any interest in such property.



Figure 6 · The five excluded instruments (s. 31(1)).

i) Why these are excluded

They are **high-value** or **high-risk** documents where the law insists on formality: cheques and notes circulate like money (Topic 5); a power of attorney hands over control of a person's affairs; a will speaks after death when the maker cannot confirm it; land transfers are the basis of property titles. In 2002 the legislature judged electronic forms too risky for them.

ii) The power to extend — s. 31(2)

The **Federal Government**, after **consultation with the Provinces**, may by **notification** apply the whole or part of the Ordinance to any of these instruments, subject to conditions.

EXAMPLE · WHAT SARA CANNOT DO ONLINE

Sara Fabrics may sign its **export contracts** electronically. But it cannot rely on the Ordinance to make an **electronic cheque** (as a negotiable instrument), an electronic **power of attorney** for Imran, or an electronic **sale deed** for its new warehouse — unless the Government has notified an extension under s. 31(2). Electronic **payments** are governed separately by the **Payment Systems and Electronic Fund Transfers Act 2007** (Topic 5).

5.6 Assessment

- **Strength.** The temporary exemptions in ss. 10–11 removed barriers to e-commerce while the systems for e-stamping and e-attestation were built.

- **Weakness.** The "whichever is later" formula left the position **open-ended** and uncertain; progress depends on Provinces and authorities acting.
- **Exclusions.** The exclusions of s. 31 are common in comparable laws, but as digital land records and e-banking grow, they may need revisiting through s. 31(2).

Table 6 · The recognition rules at a glance.

Requirement	Section	Electronic solution
Writing	4	Accessible for later reference
Original	5	Reliable integrity, legible display
Retention	6	Accessible, accurate, with origin, destination and time
Signature	7–9	Electronic signature; presumption for advanced signatures
Stamp duty	10	Not payable until provincial e-payment systems exist (minimum two years)
Attestation	11	Not required until e-attestation measures exist (minimum two years)
Certified copy	12	Printouts verified as the authority lays down
Excluded	31	Negotiable instruments, powers of attorney, trusts, wills, land contracts

5.7 Why formalities still matter

Stamp duty, attestation and notarisation exist for reasons that do not disappear online: stamp duty is **revenue** for the Provinces; attestation and notarisation provide **independent witnesses** who confirm identity and free consent, protecting weak or vulnerable parties. The Ordinance's approach — suspend these requirements for electronic documents until **electronic equivalents** exist — recognises both the needs of e-commerce and the purposes behind the formalities. The challenge is completing the electronic systems so that the temporary rules can finally give way to permanent ones.

REMEMBER · STAMP DUTY, ATTESTATION AND EXCLUSIONS IN ONE LINE

No stamp duty on electronic instruments for two years or until provincial e-payment measures, whichever is later (s. 10); no attestation or notarisation for two years or until e-attestation measures, whichever is later (s. 11); certified printouts as verified by the appropriate authority (s. 12); negotiable instruments, powers of attorney, express trusts, wills and land contracts excluded unless the Government extends the Ordinance (s. 31).

Chapter 6 · Electronic Documents — Attribution, Acknowledgment, Dispatch and Receipt

*Syllabus: "Electronic documents (attribution, acknowledgement, time and place of dispatch)". CSS 2025 Q7 (key provisions, electronic contracts). These rules decide **whose** message it is, **whether** it was received, and **when and where** a contract is made.*

6.1 Why these rules matter

IN SIMPLE WORDS · IN PLAIN WORDS

In a paper deal you know who handed you the letter and when it arrived. Online, a message may be sent by a **computer program**, by an **employee**, or by a **hacker**; it may sit unread in a **spam folder**; the sender may be in **Dubai** and the receiver in **Faisalabad**. Chapter 3 of the Ordinance answers: whose message is it, has it been acknowledged, when was it sent and received, and at what place. These answers decide when a **contract** is formed and which **court** has jurisdiction.

6.2 Attribution — s. 13

i) When the message is the originator's — s. 13(1)

Unless otherwise agreed, an electronic communication is deemed to be **that of the originator** if it was sent:

- (a) **by the originator himself**;
- (b) by a person who had **authority to act** on his behalf for that communication; or
- (c) by an **automated information system programmed by, or on behalf of,** the originator.

ii) When the addressee may rely on it — s. 13(2)

Unless otherwise agreed, the addressee may **regard** the communication as the originator's and **act on that assumption** if:

- (a) he has **no reason to suspect** its authenticity; or
- (b) there are no circumstances where he **knows, or ought to have known** by exercising **reasonable care**, that it was **not authentic**.

EXAMPLE · IMRAN'S EMAIL AND THE HACKER

(1) Imran, authorised to handle orders, emails Gulf Garments accepting a price — the acceptance is **Sara Fabrics'** (s. 13(1)(b)). (2) Sara's online store automatically confirms an order — the confirmation is Sara's (s. 13(1)(c)). (3) A hacker breaks into the email and orders 10,000 metres of silk from a supplier. If the supplier had no reason to suspect, it may treat the order as Sara's (s. 13(2)(a)); but if the email came from a strange address, at 3 a.m., with unusual terms, it **ought to have known** by reasonable care (s. 13(2)(b)) and cannot rely on it.

6.3 Acknowledgment of receipt — s. 14

- **Conditional messages — s. 14(1).** Unless otherwise agreed, where the originator has stated that the communication is **conditional on receipt of acknowledgment**, it is treated **as though it has never been sent** until the acknowledgment is received.
- **Form of acknowledgment — s. 14(2).** If no particular form or method was agreed, acknowledgment may be given by **any communication**, automated or otherwise, or by **any conduct** of the addressee sufficient to show receipt — for instance, an automatic "delivered" receipt or beginning to perform.

6.4 Time of dispatch and receipt — s. 15(1)–(3)

✦ THE LAW

Section 15(1)–(2) · *Electronic Transactions Ordinance 2002*

(1) Unless otherwise agreed …, the **dispatch** of an electronic communication occurs when it **enters an information system outside the control of the originator**. (2) Unless otherwise agreed …, or unless proved otherwise, the time of **receipt** … is determined as follows: (a) if the addressee has **designated an information system** for receiving it, receipt occurs (i) when it **enters the designated system**; or (ii) if sent to a non-designated system of the addressee, when it is **retrieved** by the addressee; (b) if the addressee has **not designated** a system, receipt occurs when it **enters an information system of the addressee**.

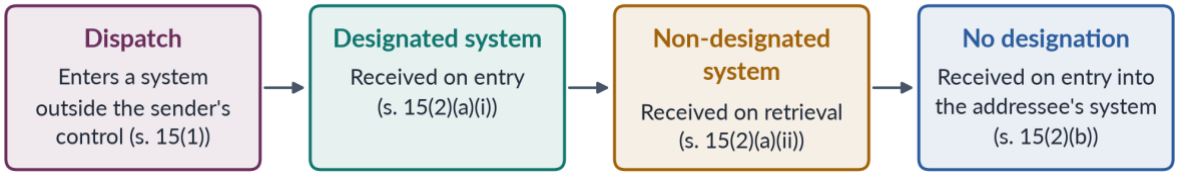


Figure 7 · When an email is sent and received (s. 15).

The time of receipt applies even if the **server** is located somewhere other than the place where the message is deemed received (s. 15(3)).

6.5 Place of dispatch and receipt — s. 15(4)–(5)

- **Rule.** Unless otherwise agreed, a communication is deemed **dispatched** where the originator **ordinarily resides or has his place of business**, and **received** where the addressee ordinarily resides or has his place of business (s. 15(4)) — **not** where the servers are.
- **Several places of business.** The one with the **closest relationship to the underlying transaction**, or else the **principal** place of business (s. 15(5)(a)).
- **No place of business.** The **usual place of residence** (s. 15(5)(b)); for a body corporate, the place where it is **incorporated** or legally constituted (s. 15(5)(c)).

EXAMPLE · WHERE AND WHEN WAS THE CONTRACT MADE?

Gulf Garments (Dubai) emails an offer; Sara Fabrics (Faisalabad) emails an acceptance from Lahore airport using Gmail, whose servers are abroad. Sara's acceptance is **dispatched** when it leaves her control (s. 15(1)) and deemed dispatched at Sara's **place of business — Faisalabad** (s. 15(4)). Gulf has designated its company email for orders, so the acceptance is **received** when it **enters** that system (s. 15(2)(a)(i)), deemed received in **Dubai**. These answers matter under the Contract Act's rules on when communication of acceptance is complete, and for which court has jurisdiction.

6.6 Electronic contracts

The Ordinance does not contain a separate section on contract formation, but its rules operate with the **Contract Act 1872** (Topic 1): an electronic offer and acceptance are valid (s. 3); the contract need not be on paper (s. 4); it may be signed electronically (s. 7); and ss. 13–15 fix **who** made the offer or acceptance and **when and where** it was communicated. **Click-wrap** agreements ("I agree") and **automated** confirmations are covered by ss. 2(1)(g), 13(1)(c).

6.7 Mistakes and automated systems

Automated systems can make mistakes — an online store may list a phone at Rs 1,000 instead of Rs 100,000 and automatically confirm thousands of orders. The Ordinance attributes the system's messages to the business that programmed it (s. 13(1)(c)), but whether a binding contract arises is decided by the **Contract Act**: an obvious pricing error that the buyer **knew or ought to have known** about may prevent a genuine agreement, and the addressee may rely on a message only where he has **no reason to suspect** it is not authentic (s. 13(2)). Businesses therefore build in **confirmation steps** and clear terms stating when a contract is formed.

REMEMBER · COMMUNICATIONS IN ONE LINE

A message is the originator's if sent by him, his agent or his programmed system (s. 13(1)); the addressee may rely on it unless he suspects or ought to know it is not authentic (s. 13(2)); a message conditional on acknowledgment is unsent until acknowledged, by any communication or conduct (s. 14); dispatch on leaving the sender's control; receipt on entry into the designated (or any) system, or on retrieval from a non-designated one (s. 15(1)–(2)); place = place of business or residence, not the server (s. 15(4)–(5)).

Chapter 7 · Certification Service Providers and the Trust Infrastructure

*Syllabus: "Certification of service providers". CSS 2025 Q7: "… the **role of certification authorities** … security, authenticity and reliability in digital transactions".*

7.1 What a certification service provider does

IN SIMPLE WORDS · IN PLAIN WORDS

How do you know that a digital signature really belongs to Ms Sara? A trusted company — a **certification service provider** — checks her identity (for example, her CNIC and company papers) and issues her a **digital certificate** linking her name to her signing key. Anyone who receives a document signed with that key can check the certificate and be confident it was Sara who signed and that the document has not changed. The provider is like a **digital notary** that vouches for identities online.

A **certificate** is "issued by a Certification Service Provider for the purpose of confirming the **authenticity or integrity** or both, of the information contained therein, of an electronic document or of an electronic signature" (s. 2(1)(h)).

7.2 How digital signatures work (public key infrastructure)

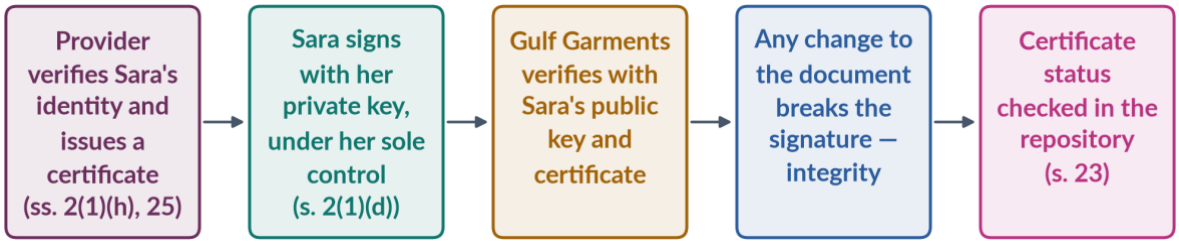


Figure 8 · Signing and verifying with a certificate.

- **Two keys.** The signer has a **private key** (secret, under his **sole control**) and a matching **public key** (shared). A document signed with the private key can be checked with the public key.
- **Integrity.** If even one character of the document changes after signing, verification **fails** — the "subsequent change … is detectable" of s. 2(1)(d)(i).
- **Identity.** The **certificate** binds the public key to the verified identity of the signer.
- **Cryptography services.** The Ordinance defines them as transforming content into a form that "cannot be understood or decoded by any unauthorized person" (s. 2(1)(k)).

7.3 Accredited and non-accredited providers — s. 17

- **Freedom to operate.** "Nothing in this Ordinance shall impede or in any way restrict the rights of any certificate service provider to engage in the business of providing certification services **without being accredited**" (s. 17(1)).
- **Protection of the title.** "No person shall **hold himself out** as an **accredited** certification service provider unless he holds a **valid accreditation certificate** issued under section 24 by the Certification Council" (s. 17(2)).

- **Why accreditation matters.** An advanced electronic signature provided by an **accredited** provider and accredited by the Council (s. 2(1)(d)(ii)), and identified in a **valid accreditation certificate**, carries the **presumption** of s. 9(a). Accreditation is voluntary but valuable.

PAKISTAN ANGLE · ACCREDITATION IN PRACTICE

The **Electronic Certification Accreditation Council (ECAC)** operates as an autonomous body under the **Ministry of Information Technology and Telecommunication**. It has accredited **NIFT** (National Institutional Facilitation Technologies), which issues certificates to **banks** for confirming the authenticity and integrity of information.

7.4 The certification practice statement — s. 25

- **Duty.** Every provider seeking accreditation must prepare and keep accessible a **certification practice statement** in the form the Council's **regulations** specify (s. 25(1)), and submit it for **approval** with its application (s. 25(3)); changes need approval (s. 25(4)); a copy is kept at the Council's office, **open to public inspection** (s. 25(5)).
- **Contents — s. 25(2).** The regulations may require: prompt information to persons adversely affected by an event or an inaccurate certificate; **identification of subscribers**; **suspension or revocation** of certificates; **accuracy** of information; foreseeability of **reliance**; and **deposit** of certificates and notices in the **repository**.
- **Statutory warranties — s. 25(6).** During the validity of an accreditation certificate published for reliance, the provider is **deemed to warrant** to any person that (a) it has **complied** with the Ordinance, rules and regulations, and (b) the information in the certificate is **accurate**.
- **Sanction — s. 25(7).** The Council may **suspend or revoke** accreditation for non-compliance, after a **reasonable right of hearing**.

7.5 Liability and offences

- **Criminal liability — s. 35.** Directors and responsible officers of a provider that issues a certificate with **false or misleading** information, fails to **revoke or suspend** a certificate it knows (or ought reasonably to know) has become false, or issues certificates as **accredited** while suspended or revoked, are punishable with up to **seven years**, a fine up to **Rs 10 million**, or both (s. 35(1)–(2)).
- **Compensation — s. 35(3)–(4).** The provider or the officers are also liable to pay **compensation** for foreseeable damage caused, recoverable as **arrears of land revenue**.
- **Subscribers — s. 34.** A subscriber who gives **false information**, fails to report **changes** making a certificate inaccurate, or knowingly allows **fraudulent** use of his certificate or signature commits an offence (up to seven years, Rs 10 million, or both).
- **Regulations — s. 43(2)(i).** The Council may provide for **reparation** to subscribers for damage arising from a provider's **negligence**, with conditions and limits.

7.6 Protection of secret keys — s. 41

"No person shall be **compelled to disclose** any password, key or other secret information **exclusively within his private knowledge**, which enables his use of the security procedure or advanced electronic signature" (s. 41(1)) — but there is **no immunity** where such information is used to commit an **offence** (s. 41(2)).

7.7 Cryptography and the telecommunications law — s. 28

Before 2002, telecom licences could restrict **cryptography**. Section 28 omitted s. 57(2)(b) of the **Pakistan Telecommunication (Re-organisation) Act 1996** and made licence terms **prohibiting cryptography services** cease to have effect, subject to the Ordinance. Section 22 makes the **Certification Council**, not the telecom regulator, **exclusively** responsible for accrediting providers, their cryptography services and security procedures.

EXAMPLE · SECURESIGN'S MISTAKE

SecureSign, an accredited provider, issues a certificate in Ms Sara's name to an impostor without checking her CNIC properly. Gulf Garments relies on a document signed with it and loses money. SecureSign is **deemed to have warranted** the accuracy of the certificate (s. 25(6)); its officers may be prosecuted if they issued it knowing, or failing to revoke it when they ought reasonably to have known, it was false (s. 35(1)); it must pay **compensation** for foreseeable damage (s. 35(3)); and the Council may **suspend** its accreditation after a hearing (s. 25(7)).

7.8 Why trust infrastructure matters

A digital signature is only as trustworthy as the **identity check** behind the certificate and the **security** of the signer's private key. If a provider issues certificates carelessly, or a signer leaves his key on an unprotected computer, the presumption of s. 9 could be used against innocent people. The Ordinance therefore combines three safeguards: **accreditation** and supervision by the Council (ss. 21–25), **criminal and civil liability** for providers and dishonest subscribers (ss. 34–35), and a **public repository** in which anyone can check whether a certificate has been suspended or revoked (s. 23).

REMEMBER · CERTIFICATION SERVICE PROVIDERS IN ONE LINE

Providers issue certificates confirming authenticity and integrity (s. 2(1)(h)); they may operate without accreditation, but none may claim to be accredited without a valid Council certificate (s. 17); accredited providers keep an approved, public certification practice statement and warrant compliance and accuracy (s. 25); false certificates are crimes with compensation (s. 35); secret keys need not be disclosed except for crimes (s. 41).

Chapter 8 · The Certification Council

*CSS 2022 Q6: "What are the **functions** of the Certification Council?" CSS 2024 Q6: "What are the **powers and functions** of the Certification Council?" CSS 2025 Q7: "… the role of certification authorities …". Syllabus: "Certification of service providers".*

8.1 Establishment — s. 18

✦ THE LAW

Section 18(1)–(2) · *Electronic Transactions Ordinance 2002*

(1) Within **sixty days** of the promulgation of this Ordinance, the **Federal Government** shall, by notification in the official Gazette, constitute a Certification Council to be known as the **Electronic Certification Accreditation Council**. (2) The Certification Council shall be a **body corporate** with **perpetual succession** and a **common seal**, and shall by the said name **sue or be sued**.

IN SIMPLE WORDS · IN PLAIN WORDS

If certification service providers are the "digital notaries", the **Certification Council** is the body that **licenses and supervises** them. It checks that a provider is technically sound and honest before accrediting it, keeps a **public register** of certificates, watches providers' compliance, and can suspend or revoke their accreditation. In practice it is known as **ECAC** and works under the Ministry of Information Technology and Telecommunication.

8.2 Composition and members — ss. 18–19



Figure 9 · The composition of the Certification Council (ss. 18–19).

- **Five members, four from the private sector**; one is designated **chairman** (s. 18(3)).
- **Qualifications — s. 19:** (a) one **telecommunications engineer** with at least **seven years'** experience, including at least **one year in cryptography services**; (b) **two** professionals or academics with at least seven years in **information technology**; (c) one with an **administrative** background and seven years' experience in a private or public organisation; (d) one **advocate** with seven years' experience and adequate knowledge of **IT and telecommunications law**.
- **Appointment and term — s. 18(4).** Appointed by the **Federal Government** for **three years**, eligible for re-appointment **once** for another three years.
- **Security of tenure — s. 18(7).** A member "shall not be removed **except on the grounds of misconduct**".
- **No conflict — s. 18(8).** No member may have any **direct financial interest** in any business relating to **cryptography services**.
- **Decisions — s. 18(9).** By **majority**; in a tie, the **chairman has a casting vote**.
- **Delegation — s. 18(6).** The Council may delegate functions to members — **except** the grant, renewal, revocation or suspension of **accreditation**.
- **Validity — s. 18(5).** No act is invalid merely because of a **vacancy** or a **defect in constitution** discovered later.

- **Service terms — s. 18(10).** As prescribed by rules.

8.3 Funds — s. 20

The Council's funds comprise: (a) **grants** from the Federal Government; (b) **fees** for grant and renewal of accreditation certificates; (c) a fee **not exceeding ten rupees** for every certificate deposited in the **repository**; and (d) **fines**.

8.4 Functions — s. 21

✦ THE LAW

Section 21(2) · *Electronic Transactions Ordinance 2002*

The Certification Council shall — (a) **grant and renew accreditation certificates** to certification service providers, their cryptography services and security procedures; (b) **monitor and ensure compliance** by accredited certification service providers with the terms of their accreditation and **revoke or suspend** accreditation in the manner and on the grounds specified in the regulations; (c) **monitor compliance** of accredited certification service providers with the provisions of this Ordinance; (d) **establish and manage the repository**; (e) carry out **research and studies** in relation to cryptography services and obtain **public opinion**; (f) **recognize or accredit foreign** certification service providers; (g) encourage **uniformity of standards and practices**; (h) give **advice** to any person in relation to any matter covered under this Ordinance; and (i) make **recommendations** to an appropriate authority in relation to the matters covered under this Ordinance.

The Council also performs any functions "specified in this Ordinance or … prescribed" (s. 21(1)).

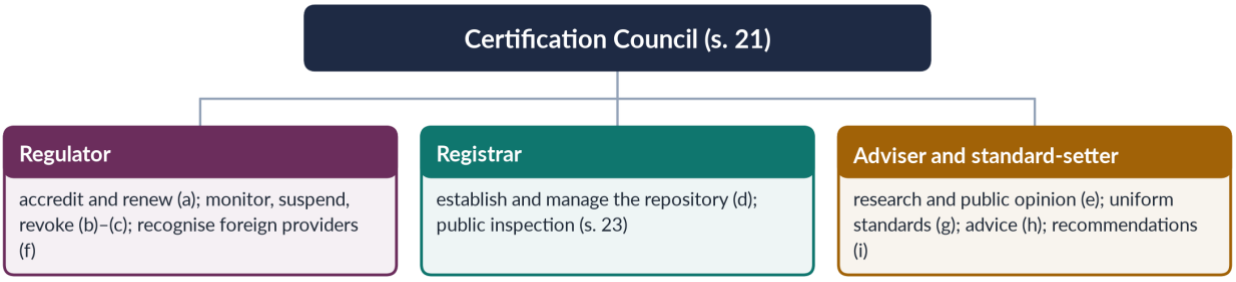


Figure 10 · The functions grouped (s. 21(2)).

8.5 Powers of the Council

i) Exclusive accreditation power — s. 22

"Notwithstanding anything contained in the Pakistan Telecommunication (Re-organisation) Act, 1996", the Council is "**exclusively responsible** to grant, renew, suspend or revoke the accreditation to certification service providers, their cryptography services and security procedures" — though telecom law still governs their **non-cryptography** telecom services (proviso).

ii) Grant of accreditation — s. 24

The Council may accredit a provider, its **cryptography services**, **electronic or advanced electronic signatures** and **security procedures** that meet the criteria in its **regulations** (s. 24(1)); terms, duration, renewal, suspension and revocation are set by regulations (s. 24(2)); fees are prescribed (s. 24(3)); and the procedure must be **transparent**, "with due regard to the **right of hearing**" (s. 24(4) proviso).

iii) Control of the practice statement — s. 25

It **approves** each provider's certification practice statement and changes to it, and may **suspend or revoke** accreditation for non-compliance after a reasonable hearing (s. 25(3)–(7)).

iv) The repository — s. 23

The Council establishes and manages a **repository** of all accreditation certificates, certificates issued by accredited providers and other specified information (s. 23(1)); it must keep it **secure** (s. 23(2)); all information is **open to public inspection** (s. 23(3)); and notices of **suspension or revocation** must be posted within the prescribed time (s. 23(4)). Certificates in the repository are **public documents** under QSO Art. 85(6).

v) Regulations — ss. 43–44

With the Federal Government's **prior approval**, the Council makes **regulations** on keys and passwords, time-stamping, staff qualifications, facilities, **data privacy**, inspections, **cross-certification** with foreign providers, reparation for negligence, certificate profiles, revocation lists, record retention and audit trails (s. 43). Drafts must be published in the Gazette and in an **English and an Urdu daily** at least **thirty days** in advance; the Council must record and answer every comment in a **report** published with the final regulations (s. 44).

vi) Decisions and staff — ss. 26–27

All matters are decided by a **speaking order**, as expeditiously as possible and **not later than ninety days**, except in extraordinary circumstances with recorded reasons (s. 26). The Council may appoint officers, employees and advisers and establish **regional offices** (s. 27).

Table 7 · Powers and functions of the Certification Council.

Power or function	Section	Content
Accreditation	21(2)(a), 22, 24	Grant and renew; exclusive; criteria in regulations; hearing
Supervision	21(2)(b)–(c), 25(7)	Monitor compliance; suspend or revoke
Practice statements	25	Approve and keep open to inspection
Repository	21(2)(d), 23	Establish, secure, public, post revocations
Foreign providers	21(2)(f), 43(2)(g)	Recognise, accredit, cross-certify
Standards, research, advice	21(2)(e), (g)–(i)	Uniform standards; studies; advice; recommendations
Regulations	43–44	With Government approval; 30-day public consultation
Decisions	26	Speaking order within 90 days

BALANCED VIEW · HOW EFFECTIVE IS THE COUNCIL?

Strengths: an **expert**, mostly **private-sector** body with protected tenure (s. 18(7)) and no conflicts (s. 18(8)); **exclusive** accreditation power (s. 22); **transparency** — public repository, public practice statements, public consultation on regulations, speaking orders within 90 days. **Weaknesses:** its **funding** depends on Government grants and small fees (s. 20); accreditation is **voluntary**, so the market for accredited certificates has grown slowly; and few businesses and citizens use **advanced** signatures, which limits the practical value of the presumption in s. 9. **The balance:** a well-designed regulator whose impact depends on **adoption** of certified signatures in banking, government and commerce.

8.7 How accreditation works, step by step

1. A provider applies in the **form** specified by the Council's regulations, with the **fee** prescribed (s. 24(3)–(4)).
2. It submits its **certification practice statement** for approval (s. 25(3)).
3. The Council examines whether it meets the **criteria** — staff qualifications, secure facilities, key management, subscriber identification, record-keeping (ss. 24(1), 43(2)).
4. It **hears** the applicant where required and decides by a **speaking order** within **ninety days** (ss. 24(4), 26).
5. On accreditation, the provider's certificates are deposited in the **repository**, which anyone may inspect (s. 23).
6. The Council **monitors** compliance and may **suspend or revoke** accreditation after a hearing, posting notice in the repository (ss. 21(2)(b), 23(4), 25(7)).

REMEMBER · THE CERTIFICATION COUNCIL IN ONE LINE

ECAC: a body corporate of five Government-appointed members, four private, with set IT, cryptography, administrative and legal qualifications, three-year terms renewable once, removable only for misconduct (ss. 18–19); funded by grants, fees and fines (s. 20); functions — accredit, monitor, suspend or revoke, keep the repository, research, recognise foreign providers, set standards, advise and recommend (s. 21); exclusive accreditation power (s. 22); speaking orders within 90 days (s. 26); regulations after public consultation (ss. 43–44).

Chapter 9 · Offences, Jurisdiction and Liability

*Syllabus: "Application to acts done outside Pakistan"; "Offences & Nature (false information, false certificates, damage to information system etc)". CSS 2025 Q7: "… **security** … What **challenges remain in its enforcement** …?"*

9.1 Why the Ordinance creates offences

IN SIMPLE WORDS · IN PLAIN WORDS

A digital-signature system works only if people can **trust** it. If a subscriber lies to get a certificate, a provider issues fake certificates, a hacker breaks into a computer, or someone deletes or alters data, trust collapses. The Ordinance therefore makes these acts **crimes**, with up to **seven years'** imprisonment.

9.2 The four offences

Table 8 · The offences under the Ordinance (ss. 34–37).

Section	Offence	Who	Maximum punishment
34	False information to a provider; failing to report changes; knowingly allowing fraudulent use of a certificate or signature	Subscriber	7 years, Rs 10 million, or both
35	Issuing a certificate with false information; failing to revoke or suspend; issuing as accredited while suspended	Directors and officers of a provider	7 years, Rs 10 million, or both; plus compensation
36	Unauthorised access to an information system (violation of privacy)	Any person	7 years, Rs 1 million, or both
37	Altering, deleting, generating or storing data without authority; impairing operation or access	Any person	7 years, Rs 1 million, or both

i) False information by the subscriber — s. 34

A subscriber who (a) gives a provider information he **knows to be false** or does not believe correct; (b) fails to **promptly** report a change that makes his certificate **inaccurate or misleading**; or (c) **knowingly** causes or allows his certificate or electronic signature to be used in a **fraudulent or unlawful** manner.

ii) False certificates — s. 35

Every **director, secretary and responsible officer** of a provider which (a) **issues, publishes or acknowledges** a certificate containing **false or misleading** information; (b) fails to **revoke or suspend** a certificate after **knowing** its information has become false; (c) fails to do so where it **ought reasonably** to have known; or (d) issues a certificate **as accredited** while its accreditation is **suspended or revoked**. The provider or officers must also pay **compensation** for foreseeable damage, recoverable as **arrears of land revenue** (s. 35(3)–(4)).

iii) Violation of privacy of information — s. 36

"Any person who **gains or attempts to gain access** to any information system with or without intent to acquire the information contained therein … whether or not he is aware of the nature or contents of

such information, **when he is not authorised** to gain access". The offence is complete on **unauthorised access**, even without intent to read or use the information.

iv) Damage to information system — s. 37

(1) Doing or attempting any act intending to **alter, modify, delete, remove, generate, transmit or store** information in an information system, **knowing** it is unauthorised; (2) doing or attempting any act intending to **impair the operation** of, or **prevent or hinder access** to, information in a system, knowing it is unauthorised.

9.3 Nature and trial of offences — ss. 38–39

- **Nature — s. 38.** All offences are **non-bailable, compoundable and cognizable**.
- **Court — s. 39.** "No Court inferior to the **Court of Sessions** shall try any offence under this Ordinance."

IN SIMPLE WORDS · WHAT THE THREE LABELS MEAN

Cognizable — the police may arrest without a warrant. **Non-bailable** — bail is not a right; the court decides. **Compoundable** — the parties may settle, and the case may end with the victim's consent.

9.4 Acts done outside Pakistan — s. 32

"The provisions of this Ordinance shall apply notwithstanding the matters being the subject hereof **occurring outside Pakistan**, in so far as they are **directly or indirectly connected to, or have an effect on or bearing** in relation to persons, information systems or events **within the territorial jurisdiction of Pakistan**." A hacker in another country who breaks into Sara Fabrics' server in Faisalabad commits an offence under s. 36 or s. 37; the practical difficulty is **catching and extraditing** him.

9.5 Limits on liability

- **Network service providers — s. 40.** "In the **absence of intent** to facilitate, aid or abet", a network service provider is **not** civilly or criminally liable **solely** because its telecommunication system was used for a contravention by a person **not under its direction or control**.
- **Secret keys — s. 41.** No one can be **compelled** to disclose passwords, keys or secret information exclusively within his knowledge, except where it was used to commit an **offence**.

EXAMPLE · THE HACKER AND THE SILK ORDER

An intruder logs into Sara Fabrics' email without permission (s. 36), changes the bank account number in an invoice to Gulf Garments (s. 37(1)), and Gulf pays the wrong account. The offences are cognizable and triable by the **Court of Sessions** (ss. 38–39); the email provider and the mobile network are not liable merely because their systems were used (s. 40). Gulf's question whether the altered invoice binds Sara is answered by **attribution** (s. 13): an unauthorised hacker is not Sara's agent, and Gulf may rely on the message only if it had no reason to suspect it.

9.6 The Ordinance and PECA 2016

The **Prevention of Electronic Crimes Act 2016 (PECA)** is now the main cyber-crime law. It creates a wider set of offences — **unauthorised access** (s. 3, up to three months or Rs 50,000), copying and **interference** with data and critical infrastructure (ss. 4–8), **electronic forgery** and **electronic fraud** (ss. 13–14),

unauthorised use of **identity information** (s. 16), **cyber stalking**, **spoofing** and others — and gives detailed powers of investigation, preservation of data, search warrants and **forensic** analysis (ss. 29–40). It was amended in **2023** and **2025**; the 2025 amendments created a **Social Media Protection and Regulatory Authority** and an offence of disseminating **false and fake information** (s. 26A). PECA operates "not in derogation of" the Penal Code, the Code of Criminal Procedure and the Qanun-e-Shahadat Order (s. 50).

Table 9 · The Ordinance's offences and PECA compared.

Point	Ordinance 2002	PECA 2016
Focus	Trust in certification; access and damage to systems	Full range of cyber-crimes
Unauthorised access	s. 36 — up to 7 years	s. 3 — up to 3 months (dishonest intention)
Data interference	s. 37	ss. 4–5, 6–8 (critical infrastructure)
Certification offences	ss. 34–35	Not specific
Nature	All non-bailable, compoundable, cognizable	Mostly non-cognizable, bailable, compoundable; serious ones not
Investigation	General criminal procedure	Special agency, warrants, preservation, forensics

BALANCED VIEW · ARE THE ORDINANCE'S OFFENCES STILL USEFUL?

For: they protect the **certification** system specifically (ss. 34–35), which PECA does not address in detail, and they carry heavy maxima. **Against:** the overlap with PECA on access and interference creates **inconsistent** punishments (seven years under s. 36 against three months under PECA s. 3); and the Ordinance gives no special **investigation** powers. **The balance:** the certification offences remain valuable; general hacking cases are better prosecuted under PECA, which was designed for them.

9.7 Proving the offences

Prosecuting electronic offences depends on **digital evidence**: server logs showing the time and source of access, the IP address used, device seizures and forensic reports. Under the Qanun-e-Shahadat Order as amended, logs generated by automated systems are **relevant** (Art. 46-A) and their printouts are **primary evidence** (Art. 73, Explanation 3), while **expert opinion** on the functioning of systems is admissible (Art. 59). Where the prosecution is brought under PECA, its search, preservation and forensic provisions (ss. 31–40) help secure such evidence lawfully.

REMEMBER · OFFENCES IN ONE LINE

Subscriber's false information (s. 34) and provider's false certificates (s. 35): seven years or Rs 10 million, plus compensation; unauthorised access (s. 36) and damage to information systems (s. 37): seven years or Rs 1 million; all non-bailable, compoundable, cognizable (s. 38), tried by the Court of Sessions (s. 39); applies to acts abroad affecting Pakistan (s. 32); network providers protected without intent (s. 40); PECA 2016 now the main cyber-crime law.

Chapter 10 · The Qanun-e-Shahadat Order and Electronic Evidence

*CSS 2026 Q8: "**Critically analyze the admissibility of electronic and computer-generated evidence under the Qanun-e-Shahadat Order, 1984** …" CSS 2023 Q5 ("appraisal" of an electronic document) and 2026 Q6 (evidentiary challenges).*

10.1 The problem the amendments solved

IN SIMPLE WORDS · IN PLAIN WORDS

The law of evidence was built for **paper and people**. It prefers the **original** document (primary evidence) to a copy; it requires the person who made a statement to come and say it; it proves signatures by handwriting. A computer printout is a copy of something that exists only as electrical signals; a bank's statement is "made" by a machine; an email has no handwriting. Without changes, much electronic evidence would be excluded or reduced to weak "secondary evidence". The 2002 Ordinance amended the **Qanun-e-Shahadat Order 1984 (QSO)** to fix these problems.

The Ordinance provides that "the Qanun-e-Shahadat Order, 1984 … shall be read subject to the amendments specified in the **Schedule**" (s. 29). The amendments are now part of the official text of the QSO.

10.2 The seven amendments made by the Ordinance

Table 10 · The Schedule's amendments to the Qanun-e-Shahadat Order.

Article	Amendment
2(1)(e)–(f)	"Automated", "electronic", "information", "information system", "electronic document", "electronic signature", "advanced electronic signature" and "security procedure" bear the Ordinance's meanings; "certificate" includes the Ordinance's meaning
30	Explanation: statements generated by **automated information systems** may be **attributed to the person exercising power or control** over the system
46-A	Statements in electronic documents **generated, received or recorded by an automated information system while it is in working order** are **relevant facts**
59	Expert opinion is relevant as to the **authenticity and integrity** of electronic documents and the **functioning, specifications, programming and operations** of information systems
73, Expl. 3	A **printout** or other output of an automated information system is **primary evidence**, if the system was in working order — which is **presumed** absent contrary evidence
73, Expl. 4	A printout of any other electronic document is **primary evidence** where a **security procedure** was applied when it was generated, sent, received or stored
78-A	If a person **denies** signing or generating an electronic document, the application of a **security procedure** must be **proved**
85(6)	**Certificates deposited in the repository** under the Ordinance are **public documents**

10.3 Relevance — Articles 30 and 46-A

- **Article 46-A** makes machine-generated statements **relevant facts** — a bank's computer-generated statement, an ATM log, a GPS record — provided the system was "**in working order**".
- **Article 30, Explanation** answers the question "whose statement is it?": statements generated by an automated system "**may be attributed to the person exercising power or control**" over it — the bank, the company, the account holder.

10.4 Proof of contents — Article 73, Explanations 3 and 4

✦ THE LAW

Article 73, Explanations 3 and 4 · *Qanun-e-Shahadat Order 1984 (inserted 2002)*

Explanation 3. A **printout** or other form of output of an **automated information system** shall not be denied the status of **primary evidence** solely for the reason that it was generated, sent, received or stored in electronic form if the automated information system was **in working order** at all material times and, for the purposes hereof, **in the absence of evidence to the contrary, it shall be presumed** that the automated information system was in working order at all material times. **Explanation 4.** A printout or other form of reproduction of an electronic document, other than a document mentioned in Explanation 3 above, first generated, sent, received or stored in electronic form, shall be treated as **primary evidence** where a **security procedure** was applied thereto at the time it was generated, sent, received or stored.

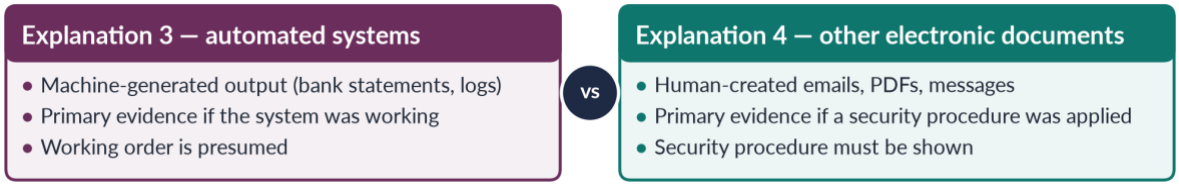


Figure 11 · Two routes to primary evidence (Article 73).

10.5 Proof of electronic signatures and documents — Article 78-A

"If an electronic document is alleged to be signed or to have been generated wholly or in part by any person through the use of an information system, and where such allegation is **denied**, the **application of a security procedure** to the signature or the electronic document **must be proved**." Together with s. 9 of the Ordinance, this gives a **two-tier** rule: an **advanced** electronic signature is **presumed** genuine; a **denied** ordinary electronic document or signature needs proof of a **security procedure** — passwords, OTPs, logs, encryption.

10.6 Experts and public documents — Articles 59 and 85(6)

- **Article 59** allows **expert opinion** on the **authenticity and integrity** of electronic documents and on the **functioning and programming** of information systems — the basis for **digital forensic** evidence.
- **Article 85(6)** makes **certificates in the repository public documents**, which may be proved by **certified copies**.

10.7 Article 164 — evidence from modern devices

i) The original text

Article 164 originally provided that "in such cases as the Court may consider appropriate, the Court may allow to be produced any evidence that may have become available because of **modern devices or techniques**." Courts used it to admit **audio and video recordings**, photographs and similar material, at their discretion.

ii) The 2023 text

The **Criminal Laws (Amendment) Act 2023** (XXXVII of 2023) substituted Article 164: "Depending on the nature of case and circumstances, the Court may, if deem appropriate, allow to be produced **any evidence or witnesses recorded by the Court through the modern devices or techniques** including **video call, viber, skype, imo, whatsapp, facebook messenger, line caller and video conference**, etc." The heading now speaks of evidence available "because of **modern devices or information system**".

iii) Its effect

The new text expressly covers **recording witnesses remotely** through modern platforms, and keeps the court's **discretion**. Article 164 does not dispense with **proof**: even admitted modern-device evidence must be shown to be **genuine** (Chapter 11).

10.8 Electronic evidence under PECA 2016

- **Forensic reports** of the investigation agency are **not inadmissible** merely because the agency produced them (PECA s. 30(7)).
- Information collected by authorised **real-time collection** "shall be **admissible** in evidence" (s. 39(2)).
- An **independent forensic laboratory** provides expert opinion on electronic evidence (s. 40).
- Seized data must be handled in a prescribed manner, and the owner may obtain a **forensic image** (s. 36).
- PECA follows the procedure of the **Code of Criminal Procedure** and the **Qanun-e-Shahadat Order** where not inconsistent (s. 44(4)).

EXAMPLE · PROVING SARA'S CASE IN COURT

Sara Fabrics sues Gulf Garments for the price. (1) Its **bank statement** showing no payment is output of an **automated system** — **primary evidence**, the system presumed working (Art. 73, Expl. 3), and attributable to the bank (Art. 30). (2) Gulf's **email** accepting the price is a human-created electronic document — primary evidence if a **security procedure** (Gulf's password-protected account) was applied (Art. 73, Expl. 4); if Gulf denies it, Sara must **prove** that security procedure (Art. 78-A). (3) The contract signed with Gulf's **SecureSign** certificate is **presumed** authentic and unaltered (s. 9). (4) A **WhatsApp voice note** from Gulf's manager may be allowed under **Article 164**, but must be proved genuine, ideally by forensic examination (Art. 59).

10.9 Hearsay and machine evidence

Ordinary evidence law treats a statement as reliable only if the person who made it testifies. A computer, however, cannot be cross-examined. The 2002 amendments solve this by treating machine records as **relevant facts** in their own right when the system was **working** (Art. 46-A) and by attributing them to the person who **controls** the system (Art. 30, Explanation). The opposing party can still challenge

the record by showing that the system was **not working** properly — which rebuts the presumption in Article 73, Explanation 3 — or that the data was **altered**.

REMEMBER · THE QANUN-E-SHAHADAT ORDER IN ONE LINE

The 2002 Schedule added electronic definitions (Art. 2), attribution of machine statements to the controller (Art. 30), relevance of automated-system records (Art. 46-A), expert opinion on electronic integrity (Art. 59), printouts as primary evidence — presumed working system or security procedure (Art. 73, Expl. 3–4), proof of a security procedure when denied (Art. 78-A) and repository certificates as public documents (Art. 85(6)); Article 164 (substituted 2023) lets courts admit evidence and record witnesses through modern devices.

Chapter 11 · Digital Evidence in the Courts

*CSS 2026 Q8: "… How does the judgment in **Alamgir Khalid Chughtai v. State (PLD 2009 Lahore 254)** contribute to the evolving jurisprudence on digital evidence in Pakistan?" CSS 2026 Q6: "… the **evidentiary challenges** that may arise in court".*

11.1 From statute to courtroom

IN SIMPLE WORDS · IN PLAIN WORDS

The Ordinance and the Qanun-e-Shahadat Order say that electronic evidence **can** be admitted. The courts decide **how much it is worth**. Because a video can be edited, a screenshot faked and an email sent from a hacked account, judges have built rules to make sure digital evidence is **genuine, complete and properly connected** to the person it is used against.

11.2 Alamgir Khalid Chughtai v State (2009)

✦ **LEADING CASE**

Alamgir Khalid Chughtai v State · *Lahore High Court, PLD 2009 Lahore 254 (2026 question)*

The Lahore High Court observed that **cyber-crime had become rampant** in society and that the **legislature had provided a different criterion for the admissibility of evidence** in such cases — recognising that electronic communications connect people across the world without any physical link. The judgment is cited in later decisions on electronic documents and the **authentication** of e-mails and electronic messages, including the point that authorship may be established in different ways — for example, where the **adverse party admits** that the texts were written by him. (The full facts of the case could not be verified for these notes, so only these points are stated.)

How the case contributes (2026)

- **Recognition of a new kind of evidence.** It was among the early High Court judgments to treat electronic evidence as governed by its **own statutory criteria** — the Ordinance's amendments to the QSO (Arts. 46-A, 73, 78-A) — rather than forcing it into paper rules.
- **Judicial awareness of cyber-crime.** By recognising that cyber-crime was **rampant**, it signalled that courts must adapt their approach to proof.
- **Authentication.** Later courts cite it on how a trial judge can be satisfied that an electronic text was really written by the other party — admission, surrounding circumstances, and proof of the account or system used.
- **A foundation for later cases.** It sits at the start of a line of authority that the Supreme Court developed into detailed **conditions** for audio, video and digital evidence (below).

11.3 Asfandyar v Kamran (2016)

✦ LEADING CASE

Asfandyar v Kamran · *Supreme Court of Pakistan, 2016 SCMR 2084*

The Supreme Court laid down conditions for relying on **audio or video tapes**: the tape cannot be relied upon unless proved **genuine and not altered or manipulated**; a **transcript** should be prepared under the supervision of an **independent person**; the **voice** of the person must be **identified**; the **source** from which it was obtained and the **date** on which it was acquired must be disclosed; the person recording it should have done so in the ordinary course rather than for personal gain or harassment; and a **formal application** to produce it must be filed.

11.4 Ishtiaq Ahmed Mirza v Federation of Pakistan (2019)

✦ LEADING CASE

Ishtiaq Ahmed Mirza v Federation of Pakistan · *Supreme Court of Pakistan, PLD 2019 SC 675*

The Supreme Court held that **no audio tape or video can be relied upon** by a court **until it is proved to be genuine** and not tampered with or doctored. Under **Article 164** it lies in the court's **discretion** to allow such evidence to be produced, but even then it must be **proved in accordance with the law of evidence**. With modern science, a **forensic examination** can establish whether a recording is genuine or has been edited; a report of the **Punjab Forensic Science Agency** on an audio or video is per se admissible under s. 9(3) of the Punjab Forensic Science Agency Act 2007. The **accuracy** of the recording must be proved, and persons who heard the conversation or saw the event may testify in support.

11.5 The principles that emerge

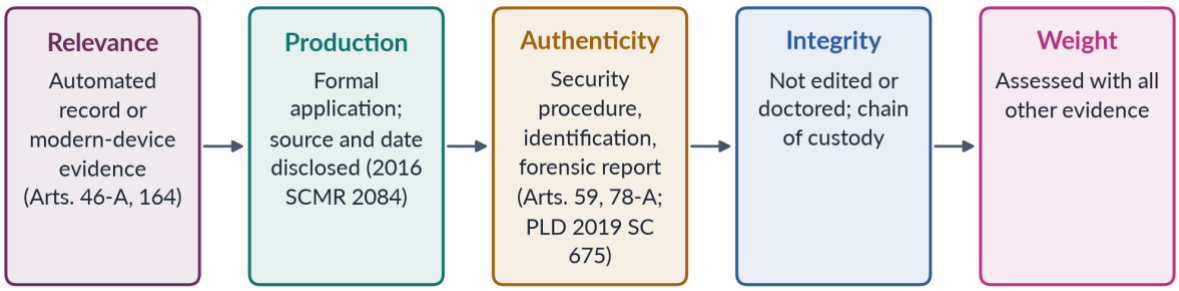


Figure 12 · Admitting a piece of digital evidence.

Table 11 · The rules courts apply to digital evidence.

Rule	Source
Electronic form alone is no ground to reject	Ordinance s. 3
Machine records relevant if the system worked	QSO Art. 46-A
Printouts are primary evidence (presumed working system, or security procedure)	QSO Art. 73, Expl. 3–4
Denied electronic document: prove the security procedure	QSO Art. 78-A

Rule	Source
Advanced signature presumed genuine	Ordinance s. 9
Modern-device evidence at the court's discretion	QSO Art. 164
Recordings must be proved genuine, untampered and accurate	PLD 2019 SC 675; 2016 SCMR 2084
Forensic expert opinion relevant	QSO Art. 59; PECA s. 40

11.6 The evidentiary challenges

i) Authenticity and tampering

Screenshots, photos, audio and video are easy to **edit**; **deepfakes** can imitate faces and voices. Courts therefore insist on **forensic** verification, and parties must preserve the **original device or file**.

ii) Chain of custody

Evidence seized from phones and computers must be handled so that it is not altered — PECA requires seized data to be dealt with in a **prescribed manner** and allows the owner a **forensic image** (s. 36).

iii) Attribution

An account may be **hacked** or **shared**. Proving that a WhatsApp message came from the accused requires more than the phone number — SIM registration, device seizure, admissions, or the circumstances (Ordinance s. 13; QSO Art. 78-A).

iv) Capacity

Many courts lack **forensic laboratories**, trained judges and technical staff; delays in forensic reports slow trials.

v) Privacy and lawfulness

Recordings made secretly, or data obtained without lawful authority, raise questions of **privacy** (Article 14 of the Constitution) and of fair trial (Article 10A). PECA's **warrant** and **preservation** procedures (ss. 31–35) are meant to ensure lawful collection.

EXAMPLE · THE WHATSAPP VOICE NOTE

In Sara Fabrics' case against Gulf Garments, Sara produces a voice note in which Gulf's manager admits the debt. Following *Asfandyar* and *Ishtiaq Ahmed Mirza*, Sara must apply formally, disclose when and how she received it, produce the **phone** that holds the original, obtain a **forensic report** that it has not been edited, and have the **voice identified**. If Gulf's manager admits sending it, authentication is much simpler — the point recognised in the case law following *Alamgir Khalid Chughtai*.

11.7 A practical checklist for digital evidence

1. **Preserve** the original device, account or file — not only screenshots.

2. **Record the source** — who obtained it, from where, and when.
3. **Apply formally** to the court to produce it, especially for audio and video.
4. **Obtain a forensic report** confirming that it is unedited and identifying metadata.
5. **Connect it to the person** — security procedure, SIM or account records, voice identification, or admission.
6. **Prepare a transcript** of recordings under independent supervision.
7. **Present printouts** of automated records with evidence that the system was working, or rely on the statutory presumption.

REMEMBER · DIGITAL EVIDENCE IN ONE LINE

Alamgir Khalid Chughtai (PLD 2009 Lahore 254): courts recognised rampant cyber-crime and a different statutory criterion for electronic evidence, and the case is cited on authentication; Asfandyar (2016 SCMR 2084): conditions for tapes — genuine, transcript, voice identified, source and date, formal application; Ishtiaq Ahmed Mirza (PLD 2019 SC 675): no recording relied on until proved genuine and untampered, Art. 164 is discretionary, forensic reports are key.

Chapter 12 · A Critical Assessment and the Digital Economy

*CSS 2025 Q7: "**Critically evaluate the effectiveness** of this Ordinance in addressing issues of **security, authenticity and reliability** in digital transactions. What **challenges remain in its enforcement**, particularly in the context of Pakistan's **evolving digital economy**?"*

12.1 How to evaluate the Ordinance

IN SIMPLE WORDS · IN PLAIN WORDS

To judge the Ordinance, ask three questions about digital deals: Are they **secure** — protected from hackers and fraud? Are they **authentic** — can we tell who made them? Are they **reliable** — can we trust that they have not changed and will stand up in court? Then ask whether the law is **working in practice** — whether people actually use its tools, and whether courts, regulators and police can enforce it.



Figure 13 · The three tests of the 2025 question.

12.2 Strengths — what the Ordinance does well

i) Legal certainty for electronic dealings

Sections 3–7 remove the old objection that electronic documents are not "writing", not "original" or not "signed". Section 30 extends paper words to electronic forms across all laws, and s. 33 gives the Ordinance **overriding effect**. Businesses can contract by email and courts must recognise it.

ii) Authenticity

- **Attribution rules** (s. 13) decide whose message it is, protecting a recipient who acts reasonably.
- A **two-tier** signature system: any electronic signature counts (s. 7), and **advanced** signatures earn a **presumption** (s. 9).
- **Accredited certification** through providers and the **Certification Council** (ss. 17–27), with a **public repository** (s. 23).

iii) Reliability

- **Integrity** is the test for originals (s. 5) and retention (s. 6).
- The **QSO amendments** make printouts **primary evidence** and require proof of **security procedures** when documents are denied (Arts. 73, 78-A).

iv) Security

- **Offences** for false information, false certificates, unauthorised access and damage to systems (ss. 34–37), with heavy maxima and trial by the **Court of Sessions**.
- Protection of **secret keys** (s. 41) and **limited liability** for network providers (s. 40).

- **Extraterritorial** reach (s. 32).

v) Forward-looking design

Technology-neutral definitions (s. 2(1)(l)) and **party autonomy** (ss. 13–15) let the Ordinance apply to platforms that did not exist in 2002 — WhatsApp, cloud storage, e-wallets.

12.3 Weaknesses — where the Ordinance falls short

i) Low use of advanced signatures

The strongest tool — the **advanced electronic signature** with its presumption — is used mainly by **banks** and institutions (for example, through **NIFT**, accredited by the Council). Most citizens and small businesses rely on emails, scans and WhatsApp, which get **no presumption** and must be proved under s. 8 and Art. 78-A.

ii) Unfinished business

- **Stamp duty and attestation** exemptions run "till" the Provinces and authorities create electronic systems (ss. 10–11) — an **open-ended** position that has depended on each Province's e-stamping arrangements.
- **Excluded instruments** — negotiable instruments, powers of attorney, wills and land contracts — remain outside unless the Government acts under s. 31(2).

iii) Overlap and gaps in security law

- The Ordinance's access and damage offences overlap with **PECA 2016**, with **inconsistent** punishments (seven years under s. 36 against three months under PECA s. 3).
- There is **no comprehensive data-protection law**: a **Personal Data Protection Bill** was approved by the Federal Cabinet but had **not been enacted** by the end of 2025, so personal data in e-commerce remains weakly protected.

iv) Institutional capacity

The Certification Council depends on **Government grants** and modest fees (s. 20); accreditation is **voluntary**; courts and police lack **forensic** capacity; and judges and lawyers need training in digital evidence.

v) Evidentiary problems

Screenshots and recordings are easy to fake; **deepfakes** are a growing risk; proving **attribution** of messages from shared or hacked accounts is hard; and the presumption that an automated system was **working** (Art. 73) favours large institutions over individuals.

vi) Cross-border enforcement

Section 32 gives jurisdiction over foreign acts with effect in Pakistan, but **investigation and extradition** depend on international cooperation, which PECA (s. 42) — not the Ordinance — addresses.

12.4 The Ordinance in Pakistan's digital economy

- **Payments.** Digital payments — **Raast** (2021), mobile wallets, internet banking — are governed by the **Payment Systems and Electronic Fund Transfers Act 2007** (Topic 5), but rest on the Ordinance's recognition of electronic records and signatures.
- **E-government.** Section 16 allows authorities to accept and issue electronic documents; e-filing of taxes, online company registration and e-court filings build on it.
- **Digital governance.** The **Digital Nation Pakistan Act 2025** created a **National Digital Commission** chaired by the Prime Minister and a **Pakistan Digital Authority** to implement a National Digital Masterplan — data governance, digital public infrastructure and digital identity.
- **Cyber-crime.** PECA, amended in **2023** and **2025** (adding a Social Media Protection and Regulatory Authority and a false-information offence), now carries most of the criminal law of cyberspace.

12.5 Reforms

Table 12 · Reforms to make the Ordinance effective.

Area	Reform
Signatures	Promote **low-cost accredited** signatures (e.g. linked to national digital identity); require them for high-value government and banking transactions
Stamp and attestation	Complete **e-stamping** and **e-attestation** systems in every Province; end the open-ended exemption
Excluded instruments	Use s. 31(2) to bring **powers of attorney** and some **land** transactions online, with safeguards
Security law	Harmonise the Ordinance's offences with **PECA**; enact a **data-protection** law
Council	Secure funding; publish **regulations** and accreditation decisions; promote **foreign** recognition
Courts	Forensic laboratories, judicial training, rules on digital **chain of custody** and **deepfake** detection

BALANCED VIEW · THE VERDICT ON THE ORDINANCE (2025)

The Ordinance is a **sound, forward-looking framework**: it gave Pakistan, from 2002, functional equivalence for electronic documents and signatures, an accreditation system, and evidence rules that let courts accept digital records. Its **weakness lies in implementation** — low uptake of accredited signatures, open-ended stamp and attestation rules, overlapping cyber-crime law, no data-protection statute, and limited forensic capacity. With the **Digital Nation Act 2025** and a growing digital economy, the priority is not a new Ordinance but **making the existing one work**: accessible certified signatures, harmonised security law, data protection and skilled courts.

12.6 A model structure for the 2025 question

1. **Introduction** — the Ordinance's purpose and the answer up front: sound framework, weak implementation.
2. **Key provisions** — recognition (ss. 3–6), signatures and presumption (ss. 7–9), stamp and attestation (ss. 10–12), communications (ss. 13–15), providers and the Council (ss. 17–27), evidence (s. 29, Schedule), offences (ss. 34–39).
3. **Evaluation** — security, authenticity and reliability, each with strengths and weaknesses.
4. **Challenges** — adoption, open-ended formalities, overlap with PECA, data protection, capacity, cross-border enforcement.

5. Reforms and conclusion — tied to the digital economy and the Digital Nation Pakistan Act 2025.

REMEMBER · THE ASSESSMENT IN ONE LINE

Strengths: legal certainty (ss. 3–7, 30, 33), authenticity through attribution, signatures and accredited certification (ss. 7–9, 13, 17–27), reliability through integrity and QSO evidence rules, security offences (ss. 34–41); weaknesses: little use of advanced signatures, open-ended stamp and attestation rules, excluded instruments, overlap with PECA, no data-protection law, weak forensic and institutional capacity; reforms: accessible certified signatures, e-stamping, harmonised cyber law, data protection, trained courts.

Part Two — Today's Debates

Short debates for the "critically evaluate", "critically examine" and "critically analyze" questions of 2025 and 2026. Each gives the issue, both sides and a line to take.

Debate 1 · Should every electronic signature be presumed genuine?

Only **advanced** electronic signatures enjoy the s. 9 presumption; ordinary electronic signatures — typed names, scans, WhatsApp messages — must be proved.

- **Extend the presumption:** most Pakistanis never use certificates; requiring proof of every email or message burdens honest parties and clogs courts.
- **Keep it narrow:** ordinary signatures are easy to forge from hacked accounts; a presumption would shift the burden onto victims of identity theft.

REMEMBER · THE LINE TO TAKE

Keep the presumption for **accredited** signatures, but make such signatures **cheap and widely available** — for example, linked to national digital identity — so that more transactions earn it.

Debate 2 · Is Article 164 a door to unreliable evidence?

Article 164, substituted in 2023, lets courts admit evidence and record witnesses through modern devices, including WhatsApp and video calls.

- **For:** it modernises trials, saves time and cost, and lets courts hear witnesses abroad.
- **Against:** recordings can be edited and **deepfakes** are increasingly convincing; remote testimony may be coached off-camera.

REMEMBER · THE LINE TO TAKE

Article 164 is a **gateway, not a guarantee**: courts must apply the Supreme Court's conditions — genuineness, forensic proof, voice identification, source and date (2016 SCMR 2084; PLD 2019 SC 675) — before giving such evidence weight.

Debate 3 · One cyber law or two?

The Ordinance's offences (ss. 34–37) overlap with PECA 2016, with different punishments and procedures.

- **Two laws:** the Ordinance protects the **certification system** specifically; PECA covers general cyber-crime.
- **One law:** overlapping offences confuse police and prosecutors and produce inconsistent sentences.

REMEMBER · THE LINE TO TAKE

Keep the **certification offences** in the Ordinance (ss. 34–35) and move or harmonise general **access and damage** offences (ss. 36–37) with PECA.

The contemporary layer

- **Criminal Laws (Amendment) Act 2023:** Article 164 of the QSO substituted to cover evidence and witnesses through video call, WhatsApp, Skype and similar platforms.
- **PECA amendments 2023 and 2025:** a Social Media Protection and Regulatory Authority; a false-and-fake-information offence (s. 26A).
- **Digital Nation Pakistan Act 2025:** National Digital Commission and Pakistan Digital Authority.
- **Data protection:** a Personal Data Protection Bill approved by the Cabinet but not enacted by the end of 2025.

Part Three — The Examiner's Record of Topic 8

Every question, 2016–2026

Table 13 · Every question the examiner has set on T8, 2016–2026.

Year · Q	The question (as set)	Family
2022 · Q6	What are the functions of the Certification Council under the Electronic Transactions Ordinance, 2002?	The Certification Council and certification service providers
2023 · Q5	What considerations have been provided by the Electronic Transactions Ordinance, 2002 for the purpose of recognition and appraisal of an electronic document?	Recognition and presumption of electronic documents and signatures
2024 · Q6	What are the powers and functions of the Certification Council under the Electronic Transactions Ordinance, 2002?	The Certification Council and certification service providers
2024 · Q7	Write a detailed note on recognition and presumption of electronic transactions under the Electronic Transactions Ordinance, 2002.	Recognition and presumption of electronic documents and signatures
2025 · Q7	With the rise of digital communication and e-commerce, the Electronic Transactions Ordinance, 2002 has played a crucial role in shaping the legal landscape of electronic contracts and digital signatures in Pakistan. Discuss the key provisions of the Ordinance, focusing on the legal recognition of electronic records and signatures, as well as the role of certification authorities. Critically evaluate the effectiveness of this Ordinance in addressing issues of security, authenticity and reliability in digital transactions. What challenges remain in its enforcement, particularly in the context of Pakistan's evolving digital economy?	The Ordinance as a whole — e-contracts, security and enforcement; Recognition and presumption of electronic documents and signatures; The Certification Council and certification service providers
2026 · Q6	Critically examine the presumptions provided under the Electronic Transactions Ordinance, 2002 regarding the authenticity and integrity of electronic records and signatures. Analyze how these presumptions facilitate the reliability of electronic transactions and the evidentiary challenges that may arise in court.	Recognition and presumption of electronic documents and signatures
2026 · Q8	Critically analyze the admissibility of electronic and computer-generated evidence under the Qanun-e-Shahadat Order, 1984. How does the judgment in Alamgir Khalid Chughtai v. State (PLD 2009 Lahore 254) contribute to the evolving jurisprudence on digital evidence in Pakistan?	Electronic evidence under the Qanun-e-Shahadat Order

What the record reveals

- **A new favourite.** No question before 2022; since then, **seven** in five papers, including **two in 2026**.
- **Recognition and presumptions** — "considerations for recognition and appraisal" (2023), "recognition and presumption" (2024), "presumptions regarding authenticity and integrity — critically examine" (2026). Learn ss. 3–12 and the QSO amendments.
- **The Certification Council** — functions (2022), powers and functions (2024). Learn ss. 18–27 and 43–44.
- **Critical questions** — the whole Ordinance with security, authenticity, reliability and enforcement challenges (2025); electronic evidence under the QSO and *Alamgir Khalid Chughtai* (2026).
- **MCQs:** four in 2025–2026 — signatures (2025, 2026), what the Ordinance covers (2025) and its aims (2026).

- **Blind spots: attribution, acknowledgment and dispatch** (ss. 13–15); **offences** (ss. 34–39); **certification service providers** (ss. 17, 25); **excluded instruments** (s. 31); **acts outside Pakistan** (s. 32).

Five likely questions

1. "Explain the rules on attribution, acknowledgment and time and place of dispatch and receipt of electronic communications." *(Chapter 6.)*
2. "Discuss the offences under the Electronic Transactions Ordinance 2002 and compare them with the Prevention of Electronic Crimes Act 2016." *(Chapter 9.)*
3. "What is a certification service provider? Discuss its duties and liabilities under the Ordinance." *(Chapter 7.)*
4. "Which instruments are excluded from the Ordinance and why? Should the exclusions be retained?" *(Chapter 5.)*
5. "Discuss the admissibility of audio and video evidence in Pakistan in the light of Article 164 and the case law of the superior courts." *(Chapters 10–11.)*

These are not solved here; the Prediction Papers include answers to predicted questions. Plan each as an answer map for practice.

Part Four — The One-Liner Bank

One-liners in the style of the past MCQs, grouped by chapter. Where a line repeats a past MCQ, the year is given in brackets. After them come all the recovered MCQs on Topic 8 with their keys.

The one-liners

A · The Ordinance and its definitions

1. The Electronic Transactions Ordinance is **Ordinance No. LI of 2002**, dated **11 September 2002**.
2. It aims to **recognise and facilitate** electronic documents and transactions and **accredit** certification service providers — **long title** (MCQ 2026: "All of these").
3. It extends to the **whole of Pakistan** and came into force **at once** — **s. 1**.
4. It does **not** cover **intellectual property rights** in digital formats (MCQ 2025).
5. "Electronic" includes electrical, digital, magnetic, optical, **biometric**, electro-chemical, wireless or electromagnetic technology — **s. 2(1)(l)**.
6. "Electronic document" includes documents, records, information, communications or transactions in electronic form — **s. 2(1)(m)**.
7. "Information" includes text, voice, video, software and **source code** — **s. 2(1)(o)**.
8. An "information system" creates, sends, receives, stores or processes information — **s. 2(1)(p)**.
9. "Automated" means **without active human intervention** — **s. 2(1)(g)**.
10. The **originator** and the **addressee** exclude an **intermediary** — **s. 2(1)(c), (t)**.
11. "Person" includes a **waqf** and an unregistered entity — **s. 2(1)(u)**.
12. "Authenticity" means **identification of and attribution** to a person or system — **s. 2(1)(f)**.
13. "Integrity" means **not tampered with, altered or modified** since a point in time — **s. 2(1)(q)**.
14. A "security procedure" may be **agreed**, **reasonably secure in business**, or set in a **practice statement** — **s. 2(1)(x)**.
15. "Transaction" means acts creating or performing **rights and obligations** — **s. 2(1)(z)**.

B · Recognition

16. No document shall be denied recognition, admissibility, effect, validity, proof or enforceability **merely because it is electronic** — **s. 3**.
17. Lack of **attestation by a witness** is no ground to deny recognition — **s. 3**.
18. "Writing" is satisfied if the electronic form is **accessible for subsequent reference** — **s. 4**.
19. "Original" is satisfied by a **reliable assurance of integrity** and **legible display** — **s. 5(1)**.
20. Integrity means remaining **complete and unaltered**, apart from normal endorsements and changes — **s. 5(2)(a)**.
21. Reliability is judged by the **purpose** and all **relevant circumstances** — **s. 5(2)(b)**.
22. Retention requires **accessibility**, **accuracy** and the **origin, destination, date and time** — **s. 6**.

23. Paper words such as "writing", "signature", "seal" and "original" **extend to electronic forms** — **s. 30**.

24. No one can **force** a government body to go digital, but it **may** accept and issue electronic documents — **s. 16**.

C · Signatures and presumption

25. An **electronic signature** is any letters, numbers, symbols, images or characters applied **with intent to authenticate or approve** — **s. 2(1)(n)**.

26. Any electronic method applied with intent to authenticate is a valid electronic signature (MCQ 2026: "All of these").

27. An **advanced** electronic signature is **unique**, **identifying**, under **sole control** and makes later changes **detectable** — **s. 2(1)(d)(i)**.

28. A signature provided by an **accredited** provider and accredited by the Council is also advanced — **s. 2(1)(d)(ii)**.

29. A legal signature requirement is satisfied by an **electronic or advanced electronic signature** — **s. 7**.

30. An electronic signature may be **proved in any manner** — **s. 8**.

31. An advanced signature is **presumed** authentic, intended and unaltered **unless contrary evidence** is adduced — **s. 9**.

32. A digital signature under an **approved certification framework** qualifies as advanced (MCQ 2025).

33. A **denied** electronic document requires proof of a **security procedure** — **QSO Art. 78-A**.

D · Stamp duty, attestation and exclusions

34. No **stamp duty** on electronic instruments for **two years** or until provincial e-payment measures, **whichever is later** — **s. 10**.

35. No **attestation or notarisation** of electronic documents for two years or until e-attestation measures, **whichever is later** — **s. 11**.

36. **Certified copies** extend to printouts verified as the appropriate authority lays down — **s. 12**.

37. **Negotiable instruments** are excluded — **s. 31(1)(a)**.

38. **Powers of attorney** are excluded — **s. 31(1)(b)**.

39. Express **trusts** are excluded, but not constructive, implied or resulting trusts — **s. 31(1)(c)**.

40. **Wills** are excluded — **s. 31(1)(d)**.

41. Contracts for sale or conveyance of **immovable property** are excluded — **s. 31(1)(e)**.

42. The Federal Government may **extend** the Ordinance to excluded instruments after consulting the Provinces — **s. 31(2)**.

E · Electronic communications

43. A message is the originator's if sent **by him**, his **authorised agent** or his **programmed automated system** — **s. 13(1)**.

44. The addressee may rely on a message unless he **suspects** or **ought to know** it is not authentic — **s. 13(2)**.

45. A message conditional on acknowledgment is treated as **never sent** until acknowledged — **s. 14(1)**.

46. Acknowledgment may be by **any communication or conduct** — **s. 14(2)**.
47. **Dispatch** occurs when a message enters a system **outside the originator's control** — **s. 15(1)**.
48. Receipt is on **entry** into a **designated** system — **s. 15(2)(a)(i)**.
49. If sent to a **non-designated** system, receipt is on **retrieval** — **s. 15(2)(a)(ii)**.
50. With no designation, receipt is on entry into **any system of the addressee** — **s. 15(2)(b)**.
51. Place of dispatch and receipt is the **place of business or residence**, not the server — **s. 15(4)**.
52. With several places of business, the one **closest to the transaction** counts — **s. 15(5)(a)**.
53. A body corporate's residence is where it is **incorporated** — **s. 15(5)(c)**.

F · Providers and the Council

54. A provider may operate **without accreditation** — **s. 17(1)**.
55. No one may claim to be **accredited** without a **valid accreditation certificate** — **s. 17(2)**.
56. The Council is the **Electronic Certification Accreditation Council** — **s. 2(1)(i), 18(1)**.
57. It was to be constituted within **sixty days** of the Ordinance — **s. 18(1)**.
58. It is a **body corporate** with perpetual succession — **s. 18(2)**.
59. It has **five members**, **four** from the **private sector** — **s. 18(3)**.
60. Members serve **three years**, re-appointable **once** — **s. 18(4)**.
61. A member may be removed **only for misconduct** — **s. 18(7)**.
62. No member may have a **financial interest** in cryptography business — **s. 18(8)**.
63. Decisions are by **majority**; the chairman has a **casting vote** — **s. 18(9)**.
64. Accreditation decisions **cannot be delegated** — **s. 18(6)**.
65. Members: a **telecom engineer** (with cryptography), **two IT** experts, an **administrator** and an **advocate**, each with **seven years** — **s. 19**.
66. Funds: **grants**, accreditation **fees**, up to **Rs 10** per repository certificate, and **fines** — **s. 20**.
67. The Council **grants, renews, monitors, suspends and revokes** accreditation — **s. 21(2)(a)–(c)**.
68. It **establishes and manages the repository** — **s. 21(2)(d)**.
69. It may **recognise or accredit foreign** providers — **s. 21(2)(f)**.
70. It encourages **uniform standards**, gives **advice** and makes **recommendations** — **s. 21(2)(g)–(i)**.
71. Its accreditation power is **exclusive**, despite the telecom law — **s. 22**.
72. The repository is **open to public inspection** — **s. 23(3)**.
73. Accreditation procedure must respect the **right of hearing** — **s. 24(4)**.
74. Every provider seeking accreditation needs an approved **certification practice statement** — **s. 25**.
75. An accredited provider **warrants** compliance and **accuracy** of certificates — **s. 25(6)**.
76. Council decisions are by **speaking order** within **ninety days** — **s. 26**.
77. Regulations need **Government approval** and **thirty days'** prior publication in English and Urdu dailies — **ss. 43–44**.
78. The Council accredited **NIFT**, which issues certificates to banks.

G · Offences and jurisdiction

79. A subscriber's **false information** is an offence — **s. 34**.

80. A provider's officers commit an offence by issuing **false certificates** — **s. 35**.

81. Offences under ss. 34–35: up to **seven years** or **Rs 10 million** — **ss. 34(2), 35(2)**.

82. **Unauthorised access** to an information system is an offence — **s. 36**.

83. **Altering, deleting** or **impairing** an information system without authority is an offence — **s. 37**.

84. Offences under ss. 36–37: up to **seven years** or **Rs 1 million** — **ss. 36, 37(3)**.

85. All offences are **non-bailable, compoundable and cognizable** — **s. 38**.

86. Offences are tried by a court **not inferior to the Court of Sessions** — **s. 39**.

87. The Ordinance applies to acts **outside Pakistan** with effect in Pakistan — **s. 32**.

88. The Ordinance has **overriding effect** — **s. 33**.

89. A network provider is not liable **without intent** to facilitate — **s. 40**.

90. No one can be compelled to disclose his **secret key**, except for offences — **s. 41**.

H · Evidence

91. The Ordinance amended the **Qanun-e-Shahadat Order** through its **Schedule** — **s. 29**.

92. Machine-generated statements may be attributed to the **person controlling** the system — **QSO Art. 30**.

93. Automated-system records are **relevant facts** if the system was **working** — **QSO Art. 46-A**.

94. Expert opinion on electronic **authenticity and integrity** is relevant — **QSO Art. 59**.

95. A printout of an automated system is **primary evidence**; working order is **presumed** — **QSO Art. 73, Expl. 3**.

96. A printout of another electronic document is primary evidence if a **security procedure** was applied — **QSO Art. 73, Expl. 4**.

97. Repository certificates are **public documents** — **QSO Art. 85(6)**.

98. Article 164 was substituted by the **Criminal Laws (Amendment) Act 2023** to cover evidence and witnesses through **video call, WhatsApp, Skype** and similar devices.

99. No audio or video may be relied on until proved **genuine and untampered** — *Ishtiaq Ahmed Mirza*, **PLD 2019 SC 675**.

100. Tapes require a **transcript**, **voice identification**, and disclosure of **source and date** — *Asfandyar v Kamran*, **2016 SCMR 2084**.

101. *Alamgir Khalid Chughtai v State*, **PLD 2009 Lahore 254**: cyber-crime is rampant and the legislature provided a **different criterion** for admitting such evidence.

102. PECA forensic reports are not inadmissible merely because the **investigation agency** produced them — **PECA s. 30(7)**.

Every past MCQ on Topic 8

Table 14 · Every recovered MCQ on T8, with its key.

Year	Question	Key
2025 · 10	Under the Electronic Transactions Ordinance, 2002, a digital signature is required to be:	(A) Created using an approved certification framework
2025 · 11	Which is not covered under the Electronic Transactions Ordinance, 2002?	(C) Intellectual property rights in digital formats
2026 · 14	The Electronic Transactions Ordinance, 2002 primarily aims to:	(D) All of these
2026 · 19	Which is considered a valid electronic signature under the Electronic Transactions Ordinance, 2002?	(D) All of these

Part Five — The Revision Sheet

The whole of Topic 8 on a few pages, for the night before the exam. Each line makes sense on its own.

1. The Ordinance and definitions

- **Ordinance LI of 2002** (11 September 2002) — recognise and facilitate electronic documents and transactions; accredit certification service providers; whole of Pakistan; acts abroad with effect in Pakistan (s. 32); overriding (s. 33). Not intellectual property.
- **Definitions** — electronic (any electro-magnetic, digital, biometric technology); electronic document; information; information system; automated; originator and addressee (not intermediaries); authenticity (attribution); integrity (unaltered); security procedure (agreed, reasonably secure, or in a practice statement). **Electronic signature** — any electronic mark with intent to authenticate (s. 2(1)(n)). **Advanced** — unique, identifying, sole control, change-detectable, or Council-accredited (s. 2(1)(d)).

2. Recognition and presumption

- **Recognition** — no denial merely because electronic or unwitnessed (s. 3); writing = accessible for later reference (s. 4); original = reliable integrity + legible (s. 5); retention = accessible, accurate, origin/destination/time (s. 6); paper words extend (s. 30); government may go digital (s. 16).
- **Signatures** — any electronic signature satisfies the law (s. 7); proof in any manner (s. 8); **advanced signature presumed** authentic, intended and unaltered unless contrary evidence (s. 9); denied documents — prove the security procedure (QSO Art. 78-A).
- **Stamp, attestation, copies** — no stamp duty (s. 10) or attestation (s. 11) for two years or until e-systems, whichever later; certified printouts (s. 12). **Excluded** — negotiable instruments, powers of attorney, express trusts, wills, land contracts, unless extended (s. 31).

3. Communications

- **Attribution** (s. 13) — originator, agent or programmed system; addressee may rely unless he suspects or ought to know. **Acknowledgment** (s. 14) — conditional message unsent until acknowledged; any communication or conduct. **Dispatch** (s. 15) — leaves the sender's control; **receipt** — entry into designated system, retrieval from non-designated, entry into any system if none designated; **place** — place of business or residence, not the server.

4. Providers and the Council

- **Providers** — may operate unaccredited; cannot claim accreditation without a valid certificate (s. 17); practice statement approved and public; warranties of compliance and accuracy; suspension or revocation after hearing (s. 25); false certificates — seven years, Rs 10 million, compensation (s. 35).
- **Council (ECAC)** — body corporate; five members, four private; telecom engineer, two IT experts, administrator, advocate, seven years each; three-year term renewable once; removal only for misconduct; no cryptography interest; majority with casting vote; accreditation non-delegable (ss. 18–19). Funds: grants, fees, Rs 10 per certificate, fines (s. 20). Functions: accredit, renew, monitor, suspend, revoke; repository; research; foreign providers; standards; advice;

recommendations (s. 21). Exclusive accreditation (s. 22); public repository (s. 23); transparent accreditation with hearing (s. 24); speaking orders in 90 days (s. 26); regulations with consultation (ss. 43–44). NIFT accredited.

5. Offences and evidence

- **Offences** — subscriber's false information (s. 34); false certificates (s. 35); unauthorised access (s. 36); damage to systems (s. 37); seven years; non-bailable, compoundable, cognizable (s. 38); Court of Sessions (s. 39); network providers protected (s. 40); secret keys (s. 41). PECA 2016 (amended 2023, 2025) is the main cyber-crime law.
- **QSO** — Art. 2(1)(e)–(f) definitions; Art. 30 attribution to controller; Art. 46-A automated records relevant; Art. 59 experts; Art. 73 Expl. 3 (printouts primary, working order presumed) and Expl. 4 (security procedure); Art. 78-A; Art. 85(6); Art. 164 (substituted 2023: video call, WhatsApp, Skype).
- **Cases** — *Alamgir Khalid Chughtai* (PLD 2009 Lahore 254): rampant cyber-crime; different statutory criterion; authentication; *Asfandyar* (2016 SCMR 2084): tape conditions; *Ishtiaq Ahmed Mirza* (PLD 2019 SC 675): genuine and untampered, forensic proof, Art. 164 discretionary.
- **Assessment** — sound framework; weak implementation: few advanced signatures, open-ended stamp and attestation, exclusions, PECA overlap, no data-protection law (end 2025), forensic capacity; Digital Nation Pakistan Act 2025.